© 2026, The Author(s).
This is an Open Access article distributed under the terms of the Creative Commons Attribution 4.0 International License (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

FILLING THE LEGAL VACUUM: THE EUROPEAN UNION'S NORMATIVE RESPONSE TO THE CYBERSECURITY OF SPACE INFRASTRUCTURE?

Croizet Fontane Romane*

In view of the growing interdependence between terrestrial and orbital systems, the cybersecurity of space infrastructure has become a critical regulatory field, yet it remains underregulated. The existing binding international instruments, having emerged in state-centric Cold War circumstances, offer little guidance on cyber-related questions. Against this background, the paper proposes to examine how the European Union, a prominent actor in space, might respond to this legal vacuum by developing a layered regulatory framework that connects space and cybersecurity regulation. It argues that through internal legal instruments, starting with general cybersecurity law and moving towards sector-specific space regulation, and by creatively using alternative normative strategies, the Union is constructing both an internal framework and a mechanism of normative projection. Drawing on the concepts of the Brussels Effect and normative power, the analysis examines how internal regulation can acquire external influence and how a regional actor can shape global regulation in the absence of binding multilateral norms. Concluding that the Union's approach can be understood as a form of regional-to-global normative leadership, this paper remains cautious regarding the effective limits of such an ambition within a fragmented international legal order.

Keywords: EU space law; space cybersecurity; space governance; Brussels Effects; critical infrastructure

* PhD Researcher in international law and space governance at the International and European Law Department of Université Côte d'Azur, specialising in EU normative power, space security, and the regulation of emerging technologies.

TABLE OF CONTENTS

INTRODUCTION.....107

I. FILLING A REGULATORY VACUUM: THE CONVERGENCE OF CYBERSECURITY AND SPACE REGULATION IN EU LAW.....117

1. The fragmented international regulation of space cybersecurity: an opportunity for the Union's intervention?.....117

2. Integrating space and cybersecurity: the EU's progressive regulation of space cybersecurity.....128

II. BEYOND INTERNAL REGULATION: THE EU'S REGULATORY ACTIVITY AS A STRATEGIC APPROACH TO SPACE CYBERSECURITY.....145

1. Global reach and normative projection: towards a Brussels Effect for the cybersecurity of space infrastructure?.....145

2. From strategic regulation to regional norm leadership: the consolidation of a European governance layer.....159

CONCLUSION.....171

INTRODUCTION

Outer space, once a domain reserved for major powers, has become a saturated and contested environment where technical, security, and geopolitical challenges converge.1 In May of 2025, more than 12,000 active satellites were in various Earth orbits, a figure approximately three times the cumulative number of satellites launched over the entire space age as of a decade ago, according to UNOOSA.2 As of today, modern societies

1 Serge Plattard and Alan Smith, 'Reducing Vulnerabilities of Space Activities: A Call for Coordinated Leadership at the Global Level' (2021) 8 Journal of Space Safety Engineering 323.

2 For up-to-date numbers, the satellite tracking website 'Orbiting Now' provides publicly available orbital elements and predictions concerning active satellites and re-entry. 'Active

increasingly depend on critical space infrastructure, whether for telecommunications, satellite navigation, or Earth observation, both for civilian and military purposes.3 The acceleration of the New Space dynamic furthered the transformation of the space sector, driven by the rise of private actors and the multiplication of satellite constellations in low Earth orbit (LEO).4 This proliferation, combined with the strong dependence of terrestrial critical infrastructures on space-based services, including commercial services, has made the space segment a prime target for hybrid and cyber threats.5

Due to their transversal nature, space infrastructures have become both essential to the functioning of states and particularly vulnerable, potentially exposed to a growing range of risks, whether physical (collisions, debris, jamming) or non-physical (cyberattacks, interference, hacking, disinformation).6 Space systems, which rely on highly interconnected digital

Satellite TLE Data and Information' (*Orbiting Now*) <http://orbit.ing-now.com> accessed 18 February 2026; 'Online Index of Objects Launched into Outer Space' (*UNOOSA*) <https://www.unoosa.org/oosa/osoindex/search-ng.jspx?lf_id=> accessed 14 January 2026.

3 Antonio Carlo and Paola Breda, 'Impact of Space Systems Capabilities and Their Role as Critical Infrastructure' (2024) 45 International Journal of Critical Infrastructure Protection 100680; Alexandru Georgescu, Olga Bucovetchi and Unal Tatar, 'Space Systems as Critical Infrastructures' (2018) 6 FAIMA Business & Management Journal 24.

4 On the impact of private actors on the space sector see William Ferguson, Scott Shackelford and Eytan Tepper, 'Cyber Threats to Space Systems and Business Opportunities for Addressing Them' (Sirius Space Talks 2025, Toulouse, 2025); see on the issue of LEO François Baccelli and others, 'Large Satellites Constellations : Issues and Impacts' (*Report from the Academy of Sciences*, 30 March 2024); see also European Union Agency for Cybersecurity and Satconsult, 'LEO Satcom Cyber Security Assessment' (*Publications Office of the EU*, 2024); and Devanshu Jha and others, 'Safeguarding the Final Frontier: Analyzing the Legal and Technical Challenges to Mega-Constellations' (2022) 9 Journal of Space Safety Engineering 636.

5 Vijay Varadharajan and Neeraj Suri, 'Security Challenges When Space Merges with Cyberspace' (2024) 67 Space Policy 101600; James Pavur, 'Securing New Space: On Satellite Cyber-Security' (DPhil thesis, University of Oxford 2021).

6 The transversal nature of space infrastructures refers to their cross-sectoral role in supporting various essential services and infrastructures across the world, rather than

and ground-based networks, are particularly exposed to attempts to compromise command-and-control functions, intercept or exfiltrate data, or disrupt services through denial-of-service operations.7 Concomitantly, electronic attacks such as spoofing, jamming, or eavesdropping specifically target the link segment, seeking to corrupt, imitate, or unlawfully capture communications between space and ground components.8 The rise in targeting of space systems by cyber operations, ever since the shift shown during the ViaSat attack of February 2022, marks a new tendency to see space asset threatened as a general trend both during and outside of armed conflict.9

A set of structural vulnerabilities inherent to a complex contemporary space ecosystem compounds these threats. The New Space industrial model, characterised by shortened development cycles, extensive reliance on commercial off-the-shelf (COTS) hardware and software components, and the proliferation of multi-tenant models for satellite operations, has significantly expanded the attack surface of space systems.10 Moreover, the

operating within a single sector or geographical zone. On the vulnerability of space infrastructure see Stefano Zatti, 'Space and Cyber Threats' in Kai-Uwe Schrogl (ed), *Handbook of Space Security* (Springer International Publishing 2020); for a comprehensive review of the literature on cybersecurity in space, including of the threats, risks and vulnerabilities of satellites, see Georgios Kavallieratos and Sokratis Katsikas, 'An Exploratory Analysis of the Last Frontier: A Systematic Literature Review of Cybersecurity in Space' (2023) 43 International Journal of Critical Infrastructure Protection 100640.

7 For instance, the Israelian Space Agency was targeted by a DDoS attack on December 2023. See Clémence Poirier and Stefan Soesanto, 'Breaking the Final Frontier: Cyber Operations against the Space Sector' (ETH Zurich Centre for Security Studies 2025).

8 Poirier and Soesanto (n 7).

9 For an analysis of the cyber operations against space systems after the Via-Sat attack, see Clémence Poirier, 'Hacking the Cosmos: Cyber Operations against the Space Sector. A Case Study from the War in Ukraine' (*Center for Security Studies*, 2024); for a further exploration of the evolution of cyber operation in the space domain since then, see Poirier and Soesanto (n 7).

10 On the New space model, see Mikko Punnala and others, 'Navigating the Space Frontier: Insights into the Current State and Future Potential of the Space Economy' (2025) 74

increasing interconnection between space and terrestrial digital infrastructures, combined with the long operational lifecycles of satellites and the difficulty of performing in-orbit patching or system upgrades, further aggravates these vulnerabilities.11 Finally, the very nature of space systems, designed on Earth, launched into orbit, and expected to operate for long periods with limited possibilities for physical intervention, means that many satellites rely on legacy hardware and software, sometimes with outdated firmware and insufficient cybersecurity mechanisms.12 The difficulty of performing in-orbit maintenance, patching, or system upgrades over increasingly complex architectures further aggravates this situation and contributes to the persistence of known vulnerabilities over extended operational lifetimes.13

The increased reliance on vulnerable satellites for communication, navigation, and strategic capabilities has brought the cybersecurity of space

Space Policy 101710; the use of Commercial Off-The-Shelf (COTS) components, ie, non-specialised hardware and software not originally designed for high-security environments, reduces costs but also increases cyber exposure, as such components are widely accessible, often insufficiently hardened, and dependent on complex supply chains that may themselves become vectors of vulnerability. Berg Nussbaum and George Berg, 'Cybersecurity Implications of Commercial Off The Shelf (COTS) Equipment in Space Infrastructure' in Unal Tatar and others (eds), *NATO Science for Peace and Security Series - D: Information and Communication Security* (IOS Press 2020); unlike traditional satellites operating on a single mission or operated by a single user, today's satellites are often in a shared usage for several usages and/or several users, which facilitate access to a satellite's capabilities for various entities. Nikita Yadav and others, 'Orbital Shield: Rethinking Satellite Security in the Commercial Off-the-Shelf Era' (2024 Security for Space Systems (3S), Netherdlands, 2024); Brianna Bace, Yasir Gökce and Unal Tatar, 'Law in Orbit: International Legal Perspectives on Cyberattacks Targeting Space Systems' (2024) 48/4 Telecommunications Policy 102739.

11 Pavur (n 5).

12 James Pavur and Ivan Martinovic, 'Building a Launchpad for Satellite Cyber-Security Research: Lessons from 60 Years of Spaceflight' (2022) 8 Journal of Cybersecurity.

13 David Livingstone and Patricia Lewis, 'Space, the Final Frontier for Cybersecurity' [2016] Chatham House, the Royal Institute of International Affairs <https://www.chathamhouse.org/2016/09/space-final-frontier-cybersecurity> accessed 6 January 2025.

infrastructure to the forefront of legal and political concern.14 The risks affecting space systems have become both more dynamic and more complex, reflecting technological advances, the multiplication of disruptive instruments, and the growing interdependence between digital and physical infrastructures.15 However, as those systems are exposed to specific threats and are subject to particular vulnerabilities, cybersecurity solutions for space systems cannot be conceived as 'one-size-fits-all'. Any solution must instead account for the specific operational constraints, architectural heterogeneity, and supply chain dependencies of space infrastructures16. This calls for a distinct regulatory approach.17

Yet, this dynamic is far from being addressed by international regulatory frameworks. Its foundational instruments were elaborated in a context where space activities were almost exclusively conducted by states, primarily for strategic purposes, and where the technological architecture of space systems remained largely isolated, quantitatively limited, and mostly physical. The central regulatory concerns at the time were sovereignty, non-appropriation, liability for physical damage, and the prevention of the weaponisation of outer space.18 Under these conditions, cybersecurity could not reasonably emerge as a regulatory priority, as space infrastructures were neither digitally networked nor exposed to systemic cyber risks. Issues of

14 Leticia Cesari, 'Cybersecurity' in PJ Blount and Mahulena Hofmann (eds) *Elgar Concise Encyclopedia of Space Law* (Elgar 2025) 54

15 André Adelsbach, Thomas Schaefer and George Tountas, 'New Space Architectures – Connectivity and Cyber Security' in PJ Blount and Mahulena Hofmann (eds) *Space Law in a Networked World* (Brill | Nijhoff 2023).

16 Ariana Vettorel, 'Cybersecurity in New Space and the Problem of International Regulation' (2024) 49 Air and Space Law 311 <https://kluwerlawonline.com/api/Product/CitationPDFURL?file=Journals\AILA\AILA2024025.pdf> accessed 20 March 2025.

17 Vettorel (n16); James Pavur and Ivan Martinovic, 'The Cyber-ASAT: On the Impact of Cyber Weapons in Outer Space' (2019 11th International Conference on Cyber Conflict (CyCon), Estonia 2019).

18 Philippe Achilleas and Stephan Hobe, *Fifty Years of Space Law* (Centre for Studies in International Law and International Relations 2020).

digital interconnection, software dependence, or systemic cyber vulnerabilities were not only technically marginal but also conceptually absent from the regulatory imagination.19 Even as space activities progressively diversified and commercial actors entered the sector, regulatory efforts continued to focus predominantly on physical safety, spectrum management, and debris mitigation.20 The progressive digitisation of space systems and their integration into global information infrastructures therefore created a widening gap between the existing framework of space law and the evolving risk environment in which space infrastructures now operate.21

Thus, the growing complexity and interdependence of space activities highlight an urgent need for legal frameworks capable of addressing cyber threats, a need clearly identified for more than a decade now.22 Yet, the international legal framework governing outer space offers little guidance on contemporary cybersecurity challenges, such as resilience, risk mitigation, attribution of responsibility, and the role of private operators. More broadly, neither existing space law nor current multilateral processes provides a coherent normative framework for protecting space infrastructure against cyber threats.23 The main generally accepted frameworks on the matter are found in industry-level initiatives stemming from standard-

19 Manfred Lachs, 'The International Law of Outer Space' (1964) 113 *Rd C 5* .

20 Almudena Azcárate Ortega and Hellmut Lagos Koller, 'The Open-Ended Working Group on Reducing Space Threats Through Norms, Rules and Principles of Responsible Behaviours: The Journey so Far, and the Road Ahead' (2023) 48 Air and Space Law <https://kluwerlawonline.com/api/Product/CitationPDFURL?file=Journals\AILA\AILA2023029.pdf> accessed 20 March 2025.

21 Gregory Falco, 'The Vacuum of Space Cyber Security' (2018 AIAA SPACE and Astronautics Forum and Exposition, American Institute of Aeronautics and Astronautics, United-States 2018).

22 For reference, Livingstone and Lewis identified the pressing matter of an under-studied and under-regulated cybersecurity issue in space systems as early as 2016, Livingstone and Lewis (n 13).

23 On multilateral processes failing to sufficiently address the cybersecurity gap in space, see Ortega and Koller (n 20).

setting bodies that lack binding power.24 While major space powers have developed national approaches reflecting divergent strategic priorities, multilateral initiatives have so far largely avoided the issue, resulting in a fragmented regulatory landscape and a persistent governance vacuum.25 In this context, the European Union occupies a distinctive position. Endowed with competences in both the space and the cybersecurity domains, and equipped with an established regulatory capacity in the governance of digital and critical infrastructures, the Union appears structurally well placed to articulate a legal response to the cybersecurity of space infrastructures and, potentially, to project this regulatory model beyond its borders.26 As a regional actor, counting amongst its members some of the main space powers, its unique position might allow for an in-between level of regulatory action, neither purely national nor multilateral, and yet somewhat global.27

This paper thus proposes to examine the role of the European Union in the emerging field of cybersecurity for space infrastructure, at the intersection of EU law and international space law, and how its regulatory actions reflect its normative power and expose the limitations of existing international space law. It starts from the observation that, in the absence of specific binding international norms addressing cyber threats to space systems, regulatory initiatives are increasingly driven at other levels.28

---

24 See for example the recommendations to Commercial space actors emitted by the National Institute of Standards and Technology; Matthew Scholl and Theresa Suloway, 'Introduction to Cybersecurity for Commercial Satellite Operations' (National Institute of Standards and Technology (US) 2023).

25 Leticia Cesari, 'Multilateral Approaches to Space Security: The Uneven Integration of 'Cyber Threats'' (2025) 50 Air and Space Law 351.

26 Francesco Cappelletti and Vagelis Papakonstantinou, 'A Question of Strategic Legislation: Can the EU Deal with Cybersecurity Issues in Space?' (2025) 49 Telecommunications Policy 102954.

27 On European space powers and the Union's position as a space power in its own right, see Mireille Couston and others, *L'Europe puissance spatiale* (E Bruylant 1991).

28 Vettorel (n 16).

Against this background, the article addresses the following research questions. Firstly, to what extent does the existing international space law framework leave a legal gap concerning the cybersecurity of space infrastructures, and why is the European Union in a position to intervene in this field? Then, how is the Union constructing an internal regulatory framework for the cybersecurity of space infrastructures, notably through instruments such as the NIS2 Directive,29 the Cyber Resilience Act,30 the Space Strategy for Security and Defence,31 and the proposed EU Space Act?32 Building on those questions, the issue of whether this regulatory activity can be understood as a form of normative projection beyond the Union's borders is raised, and whether it might be understood as a 'Brussels Effect' in the space sector. Finally, the paper explores whether this engagement, more broadly, contributes to the emergence of a new governance layer in outer space. The article argues that, taken together, these developments show that the Union is progressively filling a legal vacuum at the intersection of cybersecurity and space governance, and that its regulatory activity in this field constitutes both a normative and strategic intervention, able to position the EU as a regional norm leader in the absence of global consensus.

29 Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union, amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148 (NIS 2 Directive) [2022] OJ L333/80

30 Regulation (EU) 2024/2847 of the European Parliament and of the Council of 23 October 2024 on horizontal cybersecurity requirements for products with digital elements and amending Regulations (EU) No 168/2013 and (EU) 2019/1020 and Directive (EU) 2020/1828 (Cyber Resilience Act) [2024] OJ L 2024/2847.

31 European Commission, 'Joint Communication to the European Parliament and the Council, European Union Space Strategy for Security and Defence' (2023) JOIN (2023) 9 final.

32 European Commission, 'Proposal for a Regulation on the Safety, Resilience and Sustainability of Space Activities in the Union', (EU Space Act) (2025) COM (2025) 335 final.

While addressing the question of the EU's potential normative projection beyond its borders, the article does not claim to provide an empirical measurement of regulatory diffusion. Instead, it adopts a structural and analytical approach inspired by the literature on the Brussels Effect, examining whether the normative configuration of the European space sector, as reflected in the design of EU regulatory instruments, could generate external effects. This assessment is therefore based on legal mechanisms such as market access conditions, participation in EU programmes, and standard-setting processes, as well as on the position of European actors within the space ecosystem. Similarly, this article does not aim to provide a comparative assessment of space cybersecurity regulation across major space powers, nor to evaluate the effectiveness or industry uptake of competing regulatory models. Rather, it adopts a deliberately EU-centered perspective, focusing on the Union's regulatory activity as an intervention in an emerging field characterised by legal uncertainty and fragmented governance. The analysis, therefore, does not assess whether a Brussels Effect has materialised in practice, but interrogates whether the European legal architecture is evolving in a way that could generate external normative influence in the field of space cybersecurity. Similarly, while the law applicable to cyber operations in armed conflict raises distinct questions, this article focuses on the preventive regulatory framework applicable in peacetime, that is, the obligations and governance structures that should govern space infrastructure security regardless of whether an armed conflict is underway.

To do so, this article relies primarily on doctrinal legal research. It is based on an analysis of relevant EU primary and secondary legislation, regulatory proposals, and policy documents pertaining to cybersecurity and space governance. Such an approach appears appropriate given the paper's descriptive and analytical objectives, which aim to map the emerging EU regulatory framework applicable to space infrastructures and to assess its normative and strategic significance. 'Space infrastructure' refers here to the technical systems enabling space activities, encompassing the space segment,

the ground segment, and the link segment. 'Cyberspace' refers to the digital environment constituted by interconnected information systems, networks, and data flows. The two domains intersect in that modern space systems are increasingly dependent on digital architectures and networked communications, making them vulnerable to cyber threats. This article focuses on that intersection treating cyberspace as the threat environment and space systems as the object of the regulatory analysis.

To explore the extent to which the European Union is shaping this new framework for the cybersecurity of space infrastructure, this article proceeds in two main parts, each corresponding to a distinct but complementary element of the argument. The first part argues that the Union is progressively filling a regulatory vacuum at the intersection of space governance and cybersecurity. It first identifies the existence and nature of this gap and explores how the EU is institutionally and legally positioned to fill it. It then analyses the content and logic of the Union's regulatory activity, showing how the EU is constructing a coherent framework through the articulation of horizontal cybersecurity instruments and emerging space-specific regulatory initiatives and observing the interplay of soft law, hybrid regulation, and standard-setting being used. The second part then advances the claim that this regulatory activity should not be understood merely as internal regulation but as a broader normative and strategic intervention. It examines, first, whether and to what extent the EU framework might produce effects beyond the Union's borders, notably through mechanisms akin to a 'Brussels Effect' in the space sector. It then analyses how this combination of regulatory constraints, standard-setting practices and participation conditions in European space programmes contributes to the emergence of a new governance layer, positioning the Union as a regional norm leader in an increasingly fragmented yet under-regulated domain.

I. FILLING A REGULATORY VACUUM: THE CONVERGENCE OF CYBERSECURITY AND SPACE REGULATION IN EU LAW

1. The fragmented international regulation of space cybersecurity: an opportunity for the Union's intervention?

The progressive digitalisation and interconnection of space infrastructures have profoundly transformed both their strategic value and the breadth of their vulnerabilities.[33] Contemporary space systems, and in particular large satellite constellations, have shifted from hardware-defined architectures toward complex software-defined architectures, permanent data links with terrestrial networks, and increasingly fragmented industrial supply chains.[34] This technological evolution, however, has not been matched by a corresponding evolution in the international legal framework governing space activities. As a result, malicious cyber operations targeting space systems currently operate in a legal grey zone, where neither their precise legal qualification nor the applicable preventive obligations are clearly established. The increasing ease and stealth of cyber operations against space infrastructure, combined with the absence of explicit legal prohibitions or dedicated regulatory regimes, makes it particularly difficult to develop both credible deterrence and effective preventive governance mechanisms.[35] This combination of exposure and legal uncertainty constitute the core of the legal gap.

This gap is first and foremost a structural one. Existing space law instruments, developed in a technological context far removed from today's realities, offer little clarity on these challenges, particularly regarding responsibility attribution, obligations of resilience and mitigation, or the role of non-state actors. The legal framework governing outer space remains

---

33 On the vulnerability of space infrastructure see Zatti (n 6) and for a comprehensive review of the literature on cybersecurity in space, see Kavallieratos and Katsikas (n 6).

34 Bace and others. (n 10) 3.

35 Cesari (n 25) 352.

largely based on the Outer Space Treaty of 1967, and its subsequent instruments were conceived in a Cold War context and designed primarily to regulate inter-state relations in space.[36] The space legal regime addresses security, with notably the obligation of States Parties to conduct space activities in accordance with international law, including the Charter of the United Nations, state's responsibility and liability regarding activities in space, as well as the due regard principle, that is, the duty to conduct space activities with regard to the corresponding interests of all other States Parties to the Treaty, and initiate consultations with each other when a planned activity might interfere with the peaceful exploration and use of outer space by other states.[37] However, it does not address cybersecurity issues, having been drafted long before the digitisation of space systems and the emergence of hybrid threats targeting critical infrastructures. The foundational instruments of space law therefore offer little purchase on the cybersecurity challenges that now pervade the operational environment of space systems.

Although the international space law framework retains all its due applicability, and its general principles can be read in ways that inform cybersecurity issue's handling, its general nature leaves numerous lacunae on

[36] Treaty on principles governing the activities of States in the exploration and use of outer space, including the moon and other celestial bodies (Outer Space Treaty) (opened for signature 27 January 1967, entered into force 10 October 1967) 610 UNTS 205 (OST); Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space (opened for signature 22 April 1968, entered into force 3 December 1968) 672 UNTS 119 (Rescue Agreement); Convention on International Liability for Damage Caused by Space Objects (opened for signature 29 March 1972, entered into force 1 September 1972) 961 UNTS 187 (Liability Convention); Convention on Registration of Objects Launched into Outer Space (opened for signature 14 January 1975, entered into force 15 September 1976) 1023 UNTS 15 (Registration Convention); Agreement Governing the Activities of States on the Moon and Other Celestial Bodies (opened for signature 18 December 1979, entered into force 11 July 1984) 1363 UNTS 3 (Moon Agreement).

[37] In order: OST, art. III; OST, art. VI and art. VII, see Bin Cheng, *Studies in International Law*, vols 604–612 (Clarendon Press Oxford 1997); OST, art. IX; see Almuneda Azcárate Ortega and Victoria Samson (eds), 'A Lexicon for Outer Space Security' (UNIDIR 2023).

specific new issues.[38] Binding instruments do not address the protection of data flows between ground stations and satellites, nor do they address obligations related to the security of software and hardware components, including supply chain vulnerabilities. No widely accepted instruments provide definitions of resilience or mitigation obligations, patch management, or cybersecurity requirements applicable throughout the operational lifecycle of space systems. Instruments are largely silent on procedures for detecting, reporting, or responding to cyber incidents affecting space infrastructure, leaving operators without clear obligations in the event of attacks. Agreements have yet to be reached on even the definitions of 'cyber operation' or on the threshold for the use of force in the context of a cyberattack.[39] The absence of specific norms further complicates issues of attribution and liability in cases of malicious cyber interference with satellite services.[40] This regulatory gap leaves space actors without clear guidance on cybersecurity requirements, complicating both preventive measures and coordinated responses in the event of an attack.

Indeed, as noted in the literature, there is currently no global, comprehensive and binding legal regime governing cyber operations as

[38] For instance, the principle of due regard contained in Article IX OST may be interpreted as applying to cyber activities affecting space systems. However, its general nature and reliance on consultation rather than specific cybersecurity obligations leave significant gaps regarding the prevention, attribution, and management of cyber incidents. Lauren Napier, 'Due Regard' in PJ Blount and Mahulena Hofmann (eds) *Elgar Concise Encyclopedia of Space Law* (Edward Elgar Publishing 2025) 77.

[39] On an attempt to define 'cyber operation' see the wording of the Tallinn Manual 2.0, Michael Schmitt, *Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations* (Cambridge University Press 2017); on the policy gaps are regarding the application of cybersecurity to space-based assets, such as the lack of global consensus on a precise definition of a use of force and on a clear threshold for what constitutes a use of force in cyberspace, see Leticia Cesari, Lisa Lacroix and Antonio Carlo, 'The Challenge of Protecting Space-Based Assets against Cyber Threats' (IAC 2020 – International Astronautical Congress, International Astronautical Federation, online 2020).

[40] Du Li, 'Legal Dilemma for Combatting Malicious Cyber Activities against Space Activities' (2023) 207 Acta Astronautica 47.

such.[41] International-universal binding rules are lacking both about cybersecurity in general and regarding the cybersecurity of space systems in particular, and even instruments such as the Tallinn Manual 2.0 merely restate existing lex lata without providing operational regulatory guidance for space infrastructures.[42] State practice remains limited, opinio juris is scarce, and the absence of clear physical boundaries in cyberspace further complicates both legal qualification and enforcement.[43] Within the United Nations, and in particular under the PAROS agenda item, cyber threats to space systems are rarely addressed substantively and systematically, in part due to attribution difficulties and in part because cyber issues are often addressed in parallel diplomatic fora.[44] Existing international and regional initiatives largely focus on ex post responsibility and liability rather than on preventive security obligations. This situation reflects both the structural reluctance of states to bind themselves through hard law in the cyber domain and the absence of a shared understanding of cybersecurity itself.[45] As a

41 See Vettorel (n 16); Edward Verco E, 'Satellites Are Cyber Insecure: We Need Regulation to Avoid a Disaster' (2021) 2 ANU Journal of Law and Technology 57; Federico Bergamasco, Roberto Cassar and Popova Rada, *Cybersecurity: Key Legal Considerations for the Aviation and Space Sectors* (Kluwer Law International BV 2020); Syed Shahzad and others, 'Cyber Resilience in Critical Space Infrastructure: A Protection Framework to Better Fuse Their Legal and Technical Governance' [2026] Space Policy 101767.

42 Unlike the Tallinn Manual 1.0 (2013), which focused on cyber warfare and the law of armed conflicts, the 2.0 version (2017) extends its analysis to cyber operations in peacetime, over a varied range of domains. However, the text remains a non-binding manual, functioning on the basis of rules for specific case, that apply to the cyber question existing principles of the applicable international law. Schmitt (n 39).

43 Bace and others. (n 10) 2.

44 UNGA Res 75(47) (13 October 2020) UN Doc A/C.1/75/L.47 'Open-ended Working Group on Developments in the Field of Information and Telecommunications in the Context of International Security Established Pursuant to General Assembly Resolution 73/27 of 5 December 2018'; UNGA Res 75(240) (4 January 2021) UN Doc A/RES/75/240 'Developments in the Field of Information and Telecommunications in the Context of International Security'.

45 Andrew Liaropoulos, 'Exploring the Complexity of Cyberspace Governance: State Sovereignty, Multistakeholderism, and Power Politics' (2016) 15/4 Journal of Information Warfare 14.

result, regulatory responses have developed in a fragmented manner, often through non-binding instruments, soft law, or transnational private standards, at the risk of proliferating uncoordinated regimes ill-suited to address a global challenge.

Despite the growing urgency of the issue, multilateral responses to cybersecurity in space remain limited.[46] Existing binding instruments fail to establish specific norms or coordination mechanisms, and efforts at the UN level have so far avoided addressing space cybersecurity issues directly.[47] Major space powers, focusing on the strategic aspect of the issue, have instead adopted divergent national approaches. For instance, the United States tends to favour security and defence-focused frameworks, while China favours autonomy and control over information sovereignty.[48] Taken together, these elements explain why space cybersecurity governance today is often described as a 'regulatory vacuum' or, at best, as a patchwork of partial, ex post and largely reactive instruments.[49] Yet, the cyber protection of space systems necessarily requires a combination of technical measures and legal instruments, and in particular a shift towards genuinely preventive

46 Cesari (n 25).

47 For example, the guidelines issued in 2016 by the Working Group on the Long-term Sustainability of Outer Space Activities under the Scientific and Technical Subcommittee of the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) included some that were relevant to the issue (such as Guidelines 9, 18 and 19), yet those were not solidified into the Guidelines for the Long-term Sustainability of Outer Space Activities; Committee on the Peaceful Uses of Outer Space, 'Report on its Fifty-Ninth Session, Held in New York' (8-17 June 2016), UN Doc A/71/20; Committee on the Peaceful Uses of Outer Space, 'Report on its Sixty-Second Session, Held in New York from' (12-21 June 2019) UN Doc A/74/20.

48 On the different approaches to cyber threats regulation at the international level, see Cesari (n 25).

49 Falco (n 21).

regulatory approaches embedded in authorisation, supervision and operational requirements.[50]

According to Article VI of the Outer Space Treaty, space activities 'shall require authorisation and continuous supervision by the appropriate State Party to the Treaty', in view of which states might establish specific procedures for licensing or authorisation of various types of space activities.[51] Before granting authorisation for a space activity, States increasingly require operators to comply with conditions intended to ensure the security and resilience of space systems.[52] These conditions, defined in national space frameworks, typically combine administrative, financial, and technical requirements.[53] Within the latter, cybersecurity considerations are gradually gaining prominence, as operators may be required not only to implement protective cyber measures but also to demonstrate that these measures are appropriate and proportionate to the mission's objectives and risk profile.[54] Compliance by satellite providers with domestic requirements is not novel, as such an approach appears consistent with existing practices in the telecommunications sector, whereby satellite service providers are required to comply with domestic regulation requirements even when they do not operate physically on one's territory, as affirmed in Article 1 of the

50 Preventive (ex-ante) law designates regulatory approaches that aim to reduce risks before harm occurs by imposing obligations of anticipation, risk management and security-by-design, rather than relying primarily on sanctions or liability after damage has materialised. By contrast, reactive (ex post) law intervenes mainly through responsibility and sanction once a violation or harm has occurred. Cesari (n 25).

51 UNGA Res 68(74) (11 December 2013) UN Doc A/RES/68/74 'Recommendations on National Legislation Relevant to the Peaceful Exploration and Use of Outer Space'.

52 Cesari (n 25).

53 See most notably, United States, Commercial Space Launch Competitiveness Act 2015; France, Loi n° 2008-518 relative aux opérations spatiales 2008; Luxembourg, Loi portant sur les activités spatiales 2020; Australia, Australia Space (Launches and Returns) Act 2018; India, Norms, Guidelines and Procedures for Implementation of Indian Space Policy-2023 in respect of Authorization of Space Activities (NGP) 2024; United Kingdom, Space Industry Act 2018.

54 Cesari (n 25) 364.

International Telecommunication Union (ITU) Constitution.[55] Yet, implementing cybersecurity measures through these procedures remains a challenge.[56] Therefore, regarding cybersecurity requirements for space systems, while authorisation and supervision procedures constitute the main vector through which space law is operationalised at the national level, existing scholarship shows that, where they exist, cybersecurity requirements in the space context remain uneven, loosely defined, and largely dependent on non-binding standards or operator self-regulation.[57] In other words, the preventive aspect of cybersecurity for space systems, that is, licensing, is where the gap becomes most visible.

In this context, the European Union (EU) occupies a key position to address these issues, both as a regulator and as a normative actor, as the 'EU's unique characteristics of shared competencies in a shared domain (ie, space) offer unique perspectives on the Union's potential to lead in establishing international standards for space cybersecurity'.[58] This opportunity appears significant given the nature of the gap identified: existing international instruments largely address responsibility for space activities ex post and fail to establish preventive cybersecurity obligations, leaving the resilience of space infrastructure without adequate normative grounding at the international level.[59] Relying on legal foundations in both the space and cyber domains, the EU retains a unique opportunity to shape the normative response to the cybersecurity of space infrastructure. Indeed, the EU Treaties confer competences to the Union in both specific and wide-ranging areas,

---

55 Constitution and Convention of the International Telecommunication Union (adopted and entered into force 22 December 1992) UNTS 1825, 1826.

56 Berna Akcali Gur and Joanna Kulesza, 'The EU's Dual Policy Dilemma: Orbital Sustainability and Digital Autonomy Policies Intertwined' (2025) Space Policy 101731, 7 <https://www.sciencedirect.com/science/article/pii/S0265964625000554> accessed 19 January 2026.

57 Cesari (n 25) 364.

58 Cappelletti and Papakonstantinou (n 26).

59 Shahzad (n 41).

enabling it to act regarding Global Spaces and pursue external action objectives.[60]

Regarding space matters, the EU has two distinct legal foundations. Internally, under the title related to research and technological development and space, the EU possesses the ability to take measures in space matters. Article 189 of the Treaty on the Functioning of the European Union (TFEU) grants the Union a supporting competence in space matters, enabling it to develop its own space policy and adopt common regulations.[61] This competence remains shared between the Union and its Member States. That is, 'space' is a domain listed in the powers shared between them set out in Article 4(3) of the TFEU. This competence is therefore to be exercised by the Union while respecting the competences of its Member States. In this understanding, the space competence is rather unique as it is more of a 'parallel' competence than a classic shared one. Yet, thanks to the Union's creativity in using its competences, as well as some principles governing their exercise, such as 'consistency', intended to mitigate fragmentation across different competence domains, the EU appears overall able to formulate an internal space policy. The internal competence thus provides the Union with a basis to develop a space regulatory framework.

Beyond its internal competence, the EU's engagement with international space law finds support in the text of the OST itself. Although the EU is not formally a party to the 1967 Outer Space Treaty, as its non-state nature limits its ability to engage in international agreements, the text allows for the recognition of international organisations' activities in space, making it one

60 Joris Larik, 'EU Law and the Governance of Global Spaces: Ambitions, Constraints and Legal Creativity' (2023) 45 Journal of European Integration 1125, 1132.

61 Article 189 of the TFEU grants the Union a supporting competence, allowing it to define a space policy without preventing Member States from exercising their own, distinguishing this parallel competence from shared competence, which is subject to the principle of preemption. Consolidated version of the Treaty on the Functioning of the European Union Union [2012] OJ C326/1 ('TFEU').

of the first treaties to address international organisations explicitly.[62] Article VI of the Treaty recognises the international responsibility of intergovernmental organisations for space activities conducted under their authority, stating that States are internationally responsible for national activities in outer space and extending that responsibility to activities conducted through international organisations.[63] This responsibility is shared between the international organisation and its Member States. In addition, Article XIII clarifies the Treaty's application to international organisations, emphasising that they may act, yet that Member States cannot escape their responsibility by acting through such organisations, confirming that a shared responsibility exists between the international organisation and the States. It can therefore be argued that these provisions effectively legitimise the EU as an actor of international space law, capable of contributing to the development of norms beyond its internal market.64

Moreover, although the coexistence of both the Union and its Member States on the international scene can lead to complications regarding the expression of a single voice, and even without the use of the pre-emption principle, the issue of parallel representation seems to remain limited, maybe as their duty of sincere cooperation (Article 4(3) TEU) incites Members to avoid harming the common objectives.[65] Therefore, this double legal

---

62 Stephan Hobe, Bernhard Schmidt-Tedd and Kai-Uwe Schrogl, *Cologne Commentary on Space Law: Outer Space Treaty* (BWV Verlag 2017).

63 The Union enjoys a unique status in the sense that, although it is not a party to the 1967 Outer Space Treaty, it can be considered a space actor in its own right due to its ability to act in a coordinated and integrated manner on behalf of its Member States. Article VI of the Treaty, which places all space actors, including international organizations, on an equal footing in terms of responsibility, reinforces this position.

64 This evolution is considered to have been the result of a divergence of opinion between partisans of sole State's activities in space, notably the ex-USSR, and those of a possible private venture, most prominently the United States., Michael Gerhard, 'Article VI', in Hobe and others (n 62).

65 Contrary to comparable domains of shared competence such as the internal market, the Union retains a parallel competence rather than a shared one, and thus cannot see effects

foundation allows the EU not only to develop its own regulatory framework but also to participate in shaping broader, regional and multilateral governance structures.

Regarding cyber issues, most of the EU's competences stem from its internal market competence, as set out in Article 114 of the TFEU. Since its very first legal act in the area, the EU has developed its regulatory capacity and expanded its competences, shifting from coordination to legislative acts.66 If harmonisation is possible in the field of cybersecurity, it is mostly due to the power to facilitate the digital market (Article 114 TFEU). In cyber-related issues, the Union's competence also overlaps with other areas, including product safety law, digital services, data governance, and the more recent AI question, offering a broader range of legal bases for action. Finally, cybersecurity issues have evolved beyond secondary legislation. They are slowly gaining ground in the EU's constitutional framework, most notably in the protection of fundamental rights related to the security of personal data.[67] Yet, similarly to space matters, cybersecurity issues fall within the limits of the Union's competence, as national security remains the sole responsibility of its Member States (Article 4(2) TEU). Therefore, the boundary between cybersecurity issues, which relate to the Union's competence, and national security issues, which are retained by the Member

of the principal of pre-emption limiting Member State's ability to act internationally in shaping a common single voice in the international scene. Therefore, while the Member States themselves are Parties to the Treaties and are represented in the UN processes, whereas the Union is not in the way a State can be, there is usually an understanding that both Member States and the Union's interests align on the matter.

66 Council Decision 92/242/EEC of 31 March 1992 in the Field of Security of Information Systems, [1992] OJ L123/19; Lee Bygrave , 'The Emergence of EU Cybersecurity Law: A Tale of Lemons, Angst, Turf, Surf and Grey Boxes' (2025) 56 Computer Law & Security Review 106071.

67 See the jurisprudence of the Court of Justice of the EU with Joined Cases C-293/12 and C-594/12, *Digital Rights Ireland Ltd v. Minister for Communications, Marine and Natural Resources and Others and Kärntner Landesregierung and Others* EU:C:2014:238, as well as the jurisprudence of the European Court of Human Rights with *I v. Finland* (Merits) (Judgment) ECtHR App No 20511/03 (17 July 2008) [47].

States' competence, is blurred. Cybersecurity matters thus appear to have a legal basis for discussion at the Union's level, if only through the overlap between cybersecurity and other fields of competence, yet its action integrates into a complex and fragmented landscape.[68]

As recent comparative scholarship has shown, space cybersecurity governance in Europe and allied jurisdictions remains fragmented and non-binding, thereby creating regulatory asymmetries across the value chain, and most existing instruments apply to programme-specific systems or the ground segment.[69] Additionally, the specificity of both the space and cybersecurity domains, where international norms remain limited or fragmented, presents the EU with a window of opportunity to innovate in normative action. EU regulation in both space and cyber domains shows some proximity in methods, as both make use of internal market mechanisms and, to some extent, of the Common Security and Defence Policy. At the same time, both areas retain clear sectoral specificities, with the common limit of Member States' sole competence on national security issues. Experienced in the field of data protection, where the EU's General Data Protection Regulation (GDPR) has become a global reference, the Union now appears to be applying a similar strategy in space and cybersecurity.[70] In both fields, regulatory frameworks are still under construction, allowing

---

68 Bygrave (n 66).

69 Francesco Casaril and Letterio Galletta, 'Space Cybersecurity Governance: Assessing Policies and Frameworks in View of the Future European Space Legislation' (2025) 11/1 Journal of Cybersecurity.

70 On the topic of the data protection global effect, see Lee Bygrave, 'The 'Strasbourg Effect' on Data Protection in Light of the 'Brussels Effect': Logic, Mechanics and Prospects' (2021) 40 Computer Law & Security Review 105460; Council Decision of 31 March 1992 in the Field of Security of Information Systems 92/242/EEC [1992] OJ L123/19; on the acknowledgement by the Union of its own normative ambition in the space domain, see for instance European Commission, 'Joint Communication to the European Parliament and the Council European Union Space Strategy for Security and Defence' (2023) JOIN(2023) 9 final.

the EU to experiment with hybrid regulatory models and position itself as a normative leader.[71]

2. Integrating space and cybersecurity: the EU's progressive regulation of space cybersecurity

The EU has progressively developed a plural normative approach, incorporating space infrastructures into its main cybersecurity regulations while initiating reflection towards a dedicated framework for space cybersecurity. Although no specific legal framework is in place at the time of writing, space systems are now formally covered by several cross-sectoral instruments, marking a significant shift in how the previously exposed vulnerabilities are addressed.[72] The intersection of cybersecurity law and space law has led to the gradual inclusion of space systems within broader cybersecurity frameworks, reflecting the EU's method of layering regulations onto existing structures while progressing toward specialisation.[73] Space infrastructure, long overlooked in cybersecurity strategies, is now increasingly recognised as critical within general cybersecurity frameworks.[74] The current European approach, therefore, relies on a set of general regulations, derived from the fields of digital regulations and cybersecurity policies, within which space is increasingly treated as a critical sector.[75] Nevertheless, a fully specialised framework for space cybersecurity has yet to be formalised, despite growing political

71 Bygrave (n 66).

72 Francesco Casaril and Giovanni Tricco, 'Space Infrastructure as Critical Infrastructure: Rights Beyond Earth in the Digital Age' (2026) 2026 Technology and Regulation 116.

73 This layering consist in the succession of regulatory levels, from the general horizontal to the sector-specific. See Jean-Pierre Thil and Marie-Eve Samson (eds), *Le droit spatial saisi par le droit de l'Union européenne* (Bruylant 2025).

74 In particular, with the inclusion of space amidst the domains covered by the NIS 2 Directive (2022) (n 29), see also Casaril and Tricco (n 72).

75 See Thil and Samson (n 73).

consensus within the EU on the need to ensure the long-term security of orbital infrastructures.[76]

The progressive integration of space cybersecurity into EU regulation is not an isolated development, but rather part of a broader regulatory trajectory in which cybersecurity has gradually become a strategic priority for the Union. Recent scholarship has highlighted that the EU's growing regulatory involvement in space cybersecurity is grounded in the reclassification of space systems as critical infrastructure, whose disruption would have cascading economic, societal, and security effects.[77] This framing provides a functional justification for extending EU cybersecurity and resilience requirements to space activities. This evolution follows a typical pattern in EU policy-making: initial political recognition of the risks, followed by the definition of strategic objectives, and finally the adoption of binding regulations. The case of space infrastructure follows this pattern, in which cybersecurity concerns, initially treated as general security matters, have progressively translated into targeted regulatory initiatives.

This trajectory is reflected in a sequence of instruments that progressively embedded cybersecurity obligations in the Union's regulation. Cyber risks have increasingly been identified in EU texts, marked by the EU Cybersecurity Strategy (2020)[78] and subsequent texts such as the NIS2 Directive (2022), the Cyber Resilience Act (CRA) (2024),[79] and the Cyber

---

76 See the EU Space Act (n 32) and European Commission, 'Joint Communication to the European Parliament and the Council: The EU's Cybersecurity Strategy for the Digital Decade' (2020) JOIN (2020) 18 final.

77 Alexandru Georgescu, Clara Cotroneo and Andreaa Dinu, 'The European Union and the Protection Of Critical Space Infrastructure from Cyber-Threats: A Strategic Approach?' (2025) 24 European Conference on Cyber Warfare and Security 127.

78 Joint Communication to the European Parliament and the Council: The EU's Cybersecurity Strategy for the Digital Decade (2020) JOIN/2020/18 final.

79 Regulation (EU) 2024/2847 of the European Parliament and of the Council of 23 October 2024 on Horizontal Cybersecurity Requirements for Products with Digital Elements (Cyber Resilience Act) [2024] OJ L2024/2847.

Solidarity Act (CSoA) (2025).[80] These instruments reflect a political will to impose a cybersecurity-by-design approach, integrating digital security from the systems' conception stage. These documents build on a dynamic that emerged in the 2010s, as the Union progressively incorporated cybersecurity into its security and defence policy through key instruments such as Directive 2013/40/EU[81] and the 2014 cyber defence framework.[82] This evolution then continued with the adoption of the NIS Directive (2016) and the Cybersecurity Act (2019).[83] At the same time, space-specific initiatives, including the IRIS² programme and the EU Space Security and Defence Strategy (2023), emphasised the need to secure the entire space infrastructure, from satellites to ground control centres.[84] Taken together, these instruments trace a trajectory from political recognition to integration into horizontal regulation, allowing for the establishment of a sector-specific framework to be developed.

Recent EU instruments reflect the gradual convergence of cyber regulations with space regulations. At the time of writing, the biggest step towards incorporating space concerns into a cybersecurity framework was taken

80 Regulation (EU) 2025/38 of the European Parliament and of the Council of 19 December 2024 laying down measures to strengthen solidarity and capacities in the Union to detect, prepare for and respond to cyber threats and incidents (Cyber Solidarity Act) [2024] OJ L2025/38.

81 Directive (EU) 2013/40 of the European Parliament and of the Council of 12 August 2013 on attacks against information systems [2013] OJ L218/8.

82 Joint Communication to the European Parliament and the Council: an EU Cyber Defence Policy Framework (2022) JOIN (2022) 49 final.

83 Regulation (EU) 2019/881 of the European Parliament and of the Council of 17 April 2019 on ENISA (the European Union Agency for Cybersecurity) and on Information and Communications Technology Cybersecurity Certification (Cybersecurity Act) [2019] OJ L151/15.

84 One of the key objectives of the Secure Connectivity Programme is to 'contribute to cyber resilience by proactive and reactive defence against cyber and electromagnetic threats and operational cybersecurity'. Alexandros Kolovos, 'Strengthening Links Between European Union Space and Defence: Adopting a Combined Approach' (2023) 63 Space Policy 101534; European Commission, 'A EU Space Strategy for Security and Defence for a Stronger and More Resilient EU' (2023) JOIN (2023) 9 final.

with the NIS2 Directive, which includes satellite operators within its scope.[85] The general framework established by the NIS2 Directive addresses cybersecurity in the space sector at the Union level, at least in part. The inclusion of the space sector among the 'critical' and 'essential' sectors in the NIS 2 Directive frameworks extends binding cybersecurity and resilience obligations to both public and private operators whose activities are deemed essential, regardless of their formal classification or size.[86] Through this move, the Union does not regulate 'space' as such, but rather space as a functional component of interconnected critical infrastructures, thereby, in practice, circumventing the limitations of Article 189 TFEU.[87] NIS 2 thus represents a first meaningful convergence between cybersecurity and space in EU regulation.

However, this framework does not provide comprehensive coverage of all actors and services involved in the space sector. Although the NIS2 Directive covers certain ground-segment operators, large parts of the space sector remain outside its scope, including spacecraft, key satellite systems, micro-operators, third-country launch services and Union-owned constellations, a gap mirrored by the Critical Entities Resilience Directive's (CER) exclusion of EU Space Programme assets from its scope. Another example is that NIS2 acknowledges the vulnerability of ICT supply chains, with a mechanism for Union-level coordinated security risk assessments of critical ICT supply chains (Article 22, Recital 91), reflecting the understanding that the increasing fragmentation and globalisation of production chains multiply points of entry for malicious interference and expose structural asymmetries in security standards and technical capabilities among suppliers.[88] These gaps

85 Directive NIS 2 (2022) (n 29).

86 Directive NIS 2 (2022) (n 29).

87 Kadri Bussov, 'The European Union in International Law' (Cambridge University Press 2021) 9, 32–33.

88 Omera Khan and Daniel Sepúlveda Estay, 'Supply Chain Cyber-Resilience: Creating an Agenda for Future Research' (2015) 5 Technology Innovation Management Review 6.

confirm that NIS 2, while a significant step, remains an incomplete response to the specific needs of the cybersecurity question in the space sector.

The CER Directive (2022) also identifies the space sector as an essential sector, justifying the imposition of reinforced obligations for risk prevention. Complementarily, the European Electronic Communications Code (EECC) requires national authorities to supervise the security of services provided by satellite operators offering paid services.[89] Added to these texts is the CRA, which introduces the principles of cybersecurity by design and by default, requiring manufacturers of digital equipment, including space equipment, to integrate security requirements from the design stage throughout the product lifecycle.[90] While the application of this regulation to space products remains auxiliary, it marks a milestone in the European commitment to preventing vulnerabilities from the early technological stages. These texts are part of a movement toward the progressive codification of European cybersecurity law applied to space, creating a layered regulatory framework that extends the coverage provided by NIS 2 into new dimensions. The EU Space Act project, as a lex specialis, would reinforce this progression, as the proposed legislation aims to fill gaps left by the NIS2 and CER Directives and to cover the assets of the Union's space programme.[91]

The proposal for an EU Space Act therefore represents a significant step in the shift from general cybersecurity regulation to sector-specific rules tailored to the needs of space infrastructure. Framed as lex specialis in relation to the NIS2 Directive, this new instrument signals the EU's move toward specialisation in a domain previously addressed only through broad,

89 Directive (EU) 2018/1972 of the European Parliament and of the Council of 11 December 2018 establishing the European Electronic Communications Code [2018] OJ L321/36.

90 Car Polona and Stefano De Luca, 'EU Cyber-Resilience Act' (2024) EPRS Briefing EU Legislation in Progress <https://www.europarl.europa.eu/RegData/etudes/BRIE/2022/739259/EPRS_BRI(2022)739259_EN.pdf> accessed 5 January 2025.

91 EU Space Act (n 32).

horizontal frameworks. By positioning the proposed Space Act as a sector-specific complement to the NIS2 Directive, the EU acknowledges that the specific vulnerabilities and operational realities of space infrastructure, highlighted by the challenges outlined above, are not adequately covered by the existing legal framework and therefore would benefit from a distinct, targeted regulatory approach.[92] This development is particularly significant, as it confirms that cybersecurity regulation would no longer be transversal and that space systems would benefit from a dedicated, stricter framework – marking a shift toward specialisation in the Union's approach to space cybersecurity.

The proposed Space Act further introduces a comprehensive approach by covering all segments of space infrastructure, from ground to space, digital and physical, which reflects the recognition of space systems as integrated cyber-physical networks (Article 5). Moreover, the regulation, which would apply not only to Union-owned assets but also to national governmental and private operators, expands its scope to cover the full range of actors involved in space operations (Articles 2 and 5). This shift is further reinforced by the proposition of a bespoke resilience baseline, setting minimum cybersecurity and resilience standards specifically adapted to space infrastructure, as discussed below. In doing so, it seems the EU is seeking to address the existing regulatory vacuum to advance the progressive development of a coherent and comprehensive internal framework for the cybersecurity of space systems. Highlighting a preference for by-design and by-default logics, the EU Space Act proposal reflects a shift toward a broader approach to cybersecurity for space issues. Its references to cybersecurity throughout the lifecycle of space systems reinforce the Union's emphasis on proactively embedding security measures rather than relying solely on reactive responses. This preventive approach aligns with the broader EU strategy, as seen in cyber-related instruments such as the NIS2 Directive and the CRA,

[92] See on this positioning the exploratory memorandum of the EU Space Act proposal, especially the section 'Consistency with other Union policies', EU Space Act (n 32).

as well as related initiatives, all of which stress the importance of integrating security considerations from the earliest design phases. Such a choice seems to illustrate an oncoming evolution towards a proactive approach, specialised to the space sector yet coordinated across strategic domains.

If initially presented in a quite uncoordinated manner, the cybersecurity requirements were a significant step in the shift from general cybersecurity regulation to sector-specific rules tailored to the needs of space infrastructure. Substantively, the proposed EU Space Act goes beyond general cybersecurity duties by imposing detailed risk management and governance requirements throughout the life cycle of space systems, including obligations relating to supply chain security, business continuity (Article 87), staff training (Article 89), and crisis communication and disclosure policies (Article 90). Particularly emblematic of this preventive law turn is the introduction of a mandatory threat-led penetration testing regime (Article 88), to be carried out before launch and periodically thereafter, inspired by other sectoral resilience frameworks such as the Digital Operational Resilience Act.[93] This progression can be viewed as signalling a maturation of the Union's regulatory approach to space cybersecurity.

Furthering the process, the Danish Presidency of the Council of the European Union presented a revised version of the Act.[94] While earlier EU interventions in this field relied primarily on the indirect application of horizontal instruments such as the NIS2 Directive, this draft of the Space Act would establish a fully sector-specific and autonomous regulatory framework for space activities. This evolution is most visible in the dedicated "resilience" chapter of the draft (Chapter II), which introduces a

[93] Regulation (EU) 2022/2554 of the European Parliament and of the Council of 14 December 2022 on digital operational resilience for the financial sector [2022] OJ L333/1 (DORA).

[94] Council of the European Union, 'Compromise Version of the EU Space Act: Regulation on the Safety, Resilience and Sustainability of Space Activities in the Union (EU Space Act) - Danish Presidency Compromise Text' (2025) 2025/0335 (COD).

comprehensive set of binding obligations tailored to the specific risk profile of space systems. These provisions still explicitly position the Space Act as lex specialis with respect to the NIS2 Directive (Article 75), but also impose cybersecurity obligations on all space operators not covered by the NIS2 Directive (Article 75-a).[95] The result would therefore be a regulatory coverage of the issue that is both specific and complementary, closing the gaps left by lex generalis instruments without displacing them.

Beyond substantive obligations, the revised Act constructs an integrated supervisory and incident response architecture that integrates the specialised actors into the broader EU framework. The latest draft establishes an incident reporting architecture (Article 93), involving both EU-level and national authorities. It embeds space operators within an integrated supervisory and coordination framework centred on EUSPA. The Act would also complement the CER Directive by imposing additional resilience duties, requiring operators to conduct comprehensive risk assessments, implement security measures proportionate to the criticality of their missions, and maintain continuity protocols across all operational phases (Articles 78-88). Additionally, a new Union Space Resilience Network (EUSRN) would facilitate coordinated responses to major cyber incidents and again support harmonisation across member states (Article 94), building upon the established network of the NIS 2 Directive, that is, the computer security incident response teams network ('CSIRTs network'), ENISA and the European Cyber Crisis Liaison Organisation Network (EU-CyCLONe). Taken together, these developments appear to signal a willingness to position the notion of resilience as a structural transformation of EU space governance: space infrastructures are no longer treated merely

[95] Article 75 (a) of the Danish presidency revised version, dedicated to cybersecurity, would subject all entity that are not already concerned by the qualification of essential, important or critical and therefore subject to Directives NIS2 and CER, to a space specific set of cybersecurity obligations. The text would subject broadly space operators to risk-based cybersecurity measures following an all-hazards approach, with clarified incident reporting obligations.

as objects of technical regulation or international responsibility, but as critical cyber-physical systems subject to a fully-fledged, ex ante, compliance-oriented regime of risk management and regulatory oversight.

This dynamic appears further confirmed by the Commission's January 2026 proposal for a revised Cybersecurity Act, which explicitly identifies space services and satellite systems as part of critical ICT supply chains.[96] The proposal introduces a Union-level framework to address non-technical and structural cybersecurity risks affecting supply chains, including the phase-out of high-risk ICT components from key satellite network assets (Article 110) and the prohibition of certain suppliers in critical satellite systems (Article 111). It also seeks to strengthen institutional coordination between ENISA and EUSPA for the protection of strategic space infrastructures (Recitals 29 and 73). This evolution would continue, from a model focused on incident response and organisational risk management to one of structural risk prevention, targeting technological dependencies, design choices, and upstream vulnerabilities. In this respect, the NIS 2 Directive can be seen as the starting point of a broader process to secure critical digital and space supply chains, a process now being pursued and refined through subsequent initiatives.[97]

The preventive logic would also appear further supported by the proposal advanced on 21 January 2026 by the Commission, aimed at strengthening the regulatory framework for digital connectivity, a domain increasingly intertwined with space infrastructure.[98] The proposal for a Digital Networks

96 European Commission, 'Proposal for a Directive of the European Parliament and of the Council amending Directive (EU) 2022/2555 as regards simplification measures and alignment with the (Proposal for the Cybersecurity Act 2)' COM (2026) 11 final.

97 Akcali Gur and Kulesza (n 56) 7.

98 European Commission, 'Proposal for a Regulation of the European Parliament and of the Council on Digital Networks, Amending Regulation (EU) 2015/2120, Directive 2002/58/EC and Decision No 676/2002/EC and Repealing Regulation (EU) 2018/1971, Directive (EU) 2018/1972 and Decision No 243/2012/EU (Digital Networks Act - DNA)' COM (2026) 13 final.

Act (DNA) would aim to modernise and harmonise the Union's connectivity and electronic communications framework, replacing the existing EECC. Among its key elements, the DNA seeks to create an integrated EU market for connectivity services and introduce an EU-level spectrum authorisation framework, including for pan-European satellite communication services (Article 45). These developments reflect the Union's broader strategic agenda to embed security, resilience, and harmonised regulatory oversight into critical infrastructure for terrestrial and space-dependent digital ecosystems, thereby complementing space-specific initiatives.

Overall, although Article 189 TFEU formally excludes the harmonisation of national space policies, the Union is working toward a regulatory framework for the cybersecurity of space infrastructures by integrating space systems into its critical infrastructure and cybersecurity acquis. Within the EU Space Programme itself, security requirements already apply to Galileo, EGNOS, Copernicus, SST and GOVSATCOM

under the authority of the Security Accreditation Board (SAB), albeit only for Union flagship systems.[99] Within these parameters, the Union appears to be on the way to constructing a meaningful regulatory framework for space cybersecurity, especially in the preventive layer of said framework.

However, if the preventive layer in construction appears to be positive, the establishment of a harmonised Union-level authorisation regime for space activities by the proposed Space Act appears to be much more limited (Articles 6-13). In particular, the proposed constellation licensing

[99] Galileo (the EU global navigation satellite system), EGNOS (the European Geostationary Navigation Overlay Service), Copernicus (the EU Earth observation programme), EU Space Surveillance and Tracking (EU SST), and GOVSATCOM (the EU Governmental Satellite Communications programme) are components of the EU Space Programme; see Regulation (EU) 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme and repealing Regulations (EU) No 912/2010, (EU) No 1285/2013, (EU) No 377/2014 and Decision No 541/2014/EU [2021] OJ L170/69; Bussov (n 88) 25.

mechanism, which would allow a single authorisation for an entire constellation, is in practice subject to highly restrictive conditions, as it applies only where all satellites are identical, perform the same tasks, and are launched from the same site using the same launch vehicle (Article 9 of the Space Act), a situation which in practice rarely applies. This significantly limits the operational relevance of the simplified procedure and risks discouraging flexible deployment strategies, including the use of European launch services that are still scaling up in capacity and competitiveness. More fundamentally, the introduction of a specific derogation for large third-country constellations in Article 19, justified either by the absence of viable Union-based alternatives or by overriding strategic interests, amounts to an implicit acknowledgement of Europe's current launch and deployment dependency, notably vis-à-vis non-European providers. Finally, although the Act formally extends to third-country operators (Articles 14 to 21) and introduces an equivalence mechanism to facilitate market access (Article 16), the combination of derogations, differentiated treatment and strategic conditionality suggests that the harmonised authorisation regime may, in practice, become a new site of regulatory friction rather than a purely integrative instrument.

Complementing the use of binding legal instruments, existing or prospective, the Union relies on soft law to further its activity. Indeed, complementary to the content of the norms enacted, the form that is chosen to enact them can be said to contribute to the structuration of the European Union's normative activity. The combination of forms and strategies through which the EU designs its regulatory influence, namely soft law tools, hybrid regulatory mechanisms, and standard-setting practices, appears to be blending formal legislation with softer, industry-driven methods to address the unique challenges of space infrastructure. Soft law instruments are prominent elements of the Union's activity.[100] Such instruments, including guidelines, recommendations, and non-binding standards, offer

[100] Thil and Samson (n 73).

flexibility and adaptability, allowing the EU to engage stakeholders without the constraints of binding regulation.[101]

With this combination of soft and hard law, the EU adopts a hybrid regulatory model that combines top-down regulatory mandates with bottom-up contributions from industry actors, fostering an iterative process where practical needs inform normative development. Standardisation, in particular, serves as the primary mechanism for practical enforcement: the ECSS plays a central role in developing technical standards for space systems, covering areas such as project management, manufacturing processes, testing protocols, and launch operations.[102] While a dedicated ECSS standard does not yet cover cybersecurity, it is indirectly addressed in several documents on data and communications security.[103] In parallel, the European Telecommunications Standards Institute (ETSI) contributes by creating 'European standards' for satellite communications security, notably through its Satellite Earth Station committee.[104] Standards such as the GEO

101 Fabien Terpan, 'Soft Law in the European Union' (2015) 21(1) European Law Journal 68.

102 The list of standards can be consulted at: 'Active Standards' (*European Cooperation for Space Standardization*) <https://ecss.nl/standards/active-standards/> accessed 14 January 14 2026; see also on the relevance of regional standardisation process Katarzyna Malinowska, 'Universal Safety Standards as a Foundation of National and European Space Laws' (2025) 12 Journal of Space Safety Engineering 464.

103 For example, the ECSS-E-AS-50-25 standard adopts the TC Space Data Link protocol from the Consultative Committee for Space Data Systems (CCSDS), while integrating the Space Data Link Security (SDLS) module, which addresses the confidentiality, integrity, and authentication of space transmissions. See ECSS, 'ECSS-E-ST-10C Rev.1 – System engineering general requirements' (*ECSS*, 15 February 2017). ECSS, 'ECSS-E-ST-40C – Software' (*ECSS*, 6 March 2009), ECSS, 'ECSS-E-ST-50C Rev.1 – Communications' (*ECSS*, 1 March 2021), ECSS, 'ECSS-E-AS-50-25C Rev.1 – Adoption Notice of CCSDS 232.0-B-4, TC Space Data Link Protocol' (*ECSS*, 13 January 2023).

104 Regulation (EU) No 1025/2012 of the European Parliament and of the Council (25 October 2012) on European standardisation [2012] OJ L316/12; Regulation (EU) 2022/2480 of the European Parliament and of the Council of 14 December 2022 amending Regulation (EU) No 1025/2012 as regards decisions of European standardisation organisations concerning European standards and European standardisation deliverables [2022] OJ L323/1.

Mobile Radio (GMR), used by certain geostationary satellite operators, incorporate encryption mechanisms to protect user-to-satellite links.105 Although these standards are global in scope, they exert significant influence on the security of critical space infrastructure. In particular, the Union's proactive approach to the cybersecurity of low Earth orbit (LEO) constellations today relies primarily on these standards.[106]

Indeed, the Union's approach to space cybersecurity includes the use of alternative forms of regulation to complement hard law. In particular, standard-setting and compliance obligations are positioned as key tools within this hybrid regulatory framework. For instance, the EU Space Act proposal emphasises extending resilience obligations to subcontractors and supply chains, as implied by the regulation's coverage of all space infrastructure (Article 2 and Recital 71). While these obligations are anchored in binding legislation, their practical implementation relies heavily on technical standards and certification practices rather than on detailed prescriptive rules. This logic is visible in the Union's industrial and technological policy instruments. In its 2024 programme on 'Space Technologies for European Non-Dependence and Competitiveness', the Commission explicitly recommends the use of the standards developed by the European Space Components Coordination (ESCC) for the qualification of electrical, electronic, and electromechanical components. It refers to ECSS as the relevant framework at the equipment and system levels.[107] The same documents systematically rely on ISO standards, notably ISO 16290 on Technology Readiness Levels (TRLs), as references for assessing

105 However, these standards leave providers a certain degree of flexibility in choosing encryption algorithms and authentication schemes, which raises issues regarding uniformity and the overall level of protection.

106 European Union Agency for Cybersecurity and Satconsult (n 4).

107 European Commission, 'Space Technologies for European Non-Dependence and Competitiveness' (2024) HORIZON-CL4-2024-SPACE-01-73, 6.

technological maturity.[108] The result is a regulatory model that complement binding regulation with technical standardisation.

The reliance on harmonised standards extends beyond general recommendations to structure the development of critical and sensitive technologies, such as microelectronics and high-speed data links for space systems. The Commission explicitly links the development of radiation-hardened, high-data-rate, low-consumption components for secure space communications to formal space qualification under ECSS standards and to ISO-based TRL assessment.[109] This illustrates a certain convergence between European standards and international standardisation frameworks, and confirms that standard-setting is being used as a concrete instrument to govern the security, reliability and trustworthiness of strategic space technologies.[110] Such a reliance on harmonised industry standards by the EU is not specific to space, but rather a commonly found aspect of its digital and cyber-related frameworks, particularly apparent in the NIS 2 Directive, the Digital Service Act (DSA)[111] and the Cybersecurity Act,[112] but also of its approach to space security more broadly. Standard-setting is thus being used across diverse domain, including space, to govern the security, reliability and and trustworthiness of technologies.

Indeed, in practice, the EU has, for example, materialised this standardisation strategy through a normative 'toolbox' that regulates satellite operators' licensing regarding STM practices. While relying on non-binding guidelines, the STM toolbox integrates strong compliance incentives, including certification mechanisms comparable to those used for 5G

108 European Commission HORIZON-CL4-2024-SPACE-01-73.

109 European Commission HORIZON-CL4-2024-SPACE-01-73, 8.

110 Hans-W Micklitz, *The Role of Standards in Future EU Digital Policy Legislation* (Report Commissioned by ANEC and BEUC, 2023).

111 Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC [2022] OJ L277/1 (Digital Services Act).

112 Cappelletti and Papakonstantinou (n 26) 1.

infrastructure deployment.[113] This approach blends technical adaptation into a non-binding regulatory framework with the use of existing international principles, with the said 'toolbox' drawing, for example, on state responsibility under international law and the role of national legislations pursuant to Article 189(2) of the TFEU, which ensures that the EU's strategy remains coherent with broader legal frameworks. In this way, the EU might use standardisation both as a tool of governance and as a channel for embedding its cybersecurity norms within the operational practices of the space sector. If not yet finalised for the specific issue of space cybersecurity, this approach would enable standard-setting and compliance obligations to support the Union's normative action in this field.

Compliance also seems to serve as a central regulatory tool and strategic mechanism in the Union's approach to space governance, particularly given the absence of specific binding international frameworks and the inherent challenges of regulating private actors through traditional public law. Rather than replacing formal regulation, compliance transforms its dynamics, introducing a form of privatised norm enforcement in which private entities share responsibility for the effectiveness of norms.[114] This evolution reflects a logic of meta-regulation, characterised by structured cooperation between

---

[113] It notably includes a 'Safe Space' label, a mechanism granting privileged access to European funding, as well as the inclusion of responsible operators on a 'whitelist'; See 'The EU Toolbox for 5G Security' (*Shaping Europe's digital future,* 29 January 2020) <https://digital-strategy.ec.europa.eu/en/library/eu-toolbox-5g-security> accessed 18 February 2026.

[114] Unlike regulation, understood as an external function of public authority aimed at balancing competing objectives within a market, compliance is based on a voluntary or semi-voluntary approach, integrated into the company's own practices. It is not necessarily selfless as it serves both to protect the company and to foster its accountability, particularly in areas such as anti-corruption, environmental protection, or cybersecurity. Gaelle Hardy, 'La compliance, une privatisation de la régulation ?' (2022) Revue de Droit International d'Assas, 32 <https://univ-antilles.hal.science/hal-04072058v1> accessed 6 June 2025 ; Marie-Anne Frison-Roche, 'Le droit de la compliance, voie royale pour réguler l'espace numérique', in Pierre Bonis et Laurent Castex (eds), *Compliance et nouvelles régulations (*Annales des Mines, coll. "Enjeux numériques", 2025), 69-77 <https://mafr.fr/fr/article/le-droit-de-la-compliance-voie-royale-pour-reguler/> accessed 18 December 2025.

public authorities and industry stakeholders, where the latter become co-producers and vectors for implementing norms serving the public interest.[115] Compliance mechanisms typically take the form of codes of conduct, risk assessments, reporting obligations, and certification schemes; tools commonly used to regulate the a-territorial sectors of digital technologies or space. Within the prospective EU space sector, the Space Act would embody this logic by conditioning market access on compliance with standards on data security, orbital debris mitigation, and infrastructure resilience through a 'certification' mechanism.[116] While these requirements are firstly technical, they are also normative, as they shape private behaviour and embed public interest considerations into private operational practices. This creates a dual dynamic. A vertical, top-down structure, where EU laws and regulatory frameworks define the standards to be met, and a horizontal process whereby private actors integrate compliance measures both to meet legal obligations and to anticipate market expectations.

The governance of space cybersecurity is particularly challenging as effective protection of space systems depends on both public regulatory authority and private technical capabilities; two sets of actors with different incentive structures, accountability frameworks, and operational constraints. Effective protection of space systems cannot rely exclusively on either top-down approaches or purely voluntary industry self-regulation but requires hybrid governance arrangements combining public oversight with private implementation.[117] In this perspective, the EU's use of compliance in its regulatory frameworks might offer a particularly relevant model. By imposing legally binding risk-management obligations, security-by-design requirements, and incident-reporting duties on private operators, while

[115] Frison-Roche (n 117), 38.

[116] Such mechanism is to be found in the proposed Union Space Label, EU Space Act proposal, Article 111 and 112.

[117] Gokhan Ikitemur, Bilge Karabacak and Andy Igonor, 'A Mixed Public-Private Partnership Approach for Cyber Resilience of Space Technologies' (2020) 57 NATO Science for Peace and Security Series - D: Information and Communication Security 120.

leaving them discretion over the concrete technical means of implementation, such frameworks structure a form of public–private cooperation. Furthermore, the use of certification schemes, notification requirements, standard adoption (via ECSS or ISO), and conditions attached to public contracts illustrates how the EU delegates norm enforcement to private actors while retaining structural control over the standards they must meet to regulate space activities. These instruments appear to function as de facto normative tools, allowing the EU to project its standards in a sector marked by multi-level governance and transformed state-based regulation. That said, notwithstanding its coherence or usefulness, the EU's reliance on soft law in space cybersecurity regulation relies on voluntary industry compliance and market-driven adoption, as these instruments lack formal enforceability.[118] This non-binding nature limits the EU's ability to ensure uniform application.

To sum up, the Union's approach to space cybersecurity regulation illustrates a gradual but deliberate trajectory of recognition and specialisation. In a complex context, the EU seems to have chosen to govern space cybersecurity through a layered architecture that combines horizontal instruments, sector-specific legislation and standardisation-based compliance mechanisms. This method reflects both the Union's traditional reliance on norm diffusion and a functional response to the structural fragmentation of global space governance. Initially addressed through political and strategic initiatives, the issue was later incorporated into broad, cross-sectoral instruments. Following the recognition of space systems as a critical infrastructure, the protection is on the way to being formalised through a dedicated framework, as the proposal for the EU Space Act would mark a significant step in the formal recognition of space systems' unique vulnerabilities and operational realities. Yet, this specialisation does not occur in isolation; it integrates into a regulatory architecture in which

[118] Leticia Cesari, 'Éthique et responsabilité dans l'exploration spatiale : l'essor de la soft law pour encadrer les nouvelles pratiques' (Notes de l'Ifri, 2023).

horizontal frameworks and sector-specific rules operate simultaneously and is operationalised through soft law and standards. Above all, this architecture is structured around a preventive logic: embedding cybersecurity and resilience requirements upstream, at the design, qualification and operational planning stages, as Union seeks to move from a model centred on incident response to one based on systemic risk management. In doing so, the EU is not only strengthening the internal resilience of its space sector but also consolidating a regulatory model capable of projecting its normative preferences beyond its borders.

II. BEYOND INTERNAL REGULATION: THE EU'S REGULATORY ACTIVITY AS A STRATEGIC APPROACH TO SPACE CYBERSECURITY

1. Global reach and normative projection: towards a Brussels Effect for the cybersecurity of space infrastructure?

Beyond the internal consolidation of its regulatory framework, the European Union's approach to space cybersecurity raises questions about its potential external effects. More specifically, it invites an assessment of whether the Union is in the process of creating the conditions for a form of the 'Brussels Effect' in the space domain, namely the capacity to shape global practices through the unilateral projection of its internal regulatory standards.

The notion of the Brussels Effect, as conceptualised by Bradford, refers to the European Union's capacity to project its regulatory influence beyond its borders through market mechanisms.[119] It describes how non-EU companies adopt EU rules to maintain access to the European market, thereby extending the EU's legal influence globally without formal extraterritorial enforcement. This mechanism has been particularly visible in the digital domain, most notably with the GDPR, whose standards have become de facto global benchmarks. More broadly, this dynamic is closely connected

[119] Anu Bradford, *The Brussels Effect: How the European Union Rules the World* (Oxford University Press 2020).

to the idea of the EU as a 'normative power', that is, an actor that seeks to shape international behaviour less through coercion than through the diffusion of legal standards, regulatory models and conceptions of appropriate conduct.[120] This phenomenon could be tied to the EU's global legal reach, manifesting through a wide array of external-facing actions: from asserting EU values in international organisations and treaties to shaping foreign legal regimes, enforcing compliance beyond EU borders, and even conducting regulatory investigations in third countries.[121] This influence operates through different degrees of extraterritoriality, including direct application and more nuanced forms such as 'extraterritorial extension,' in which the EU governs conduct beyond its territory.[122]

Applied to space cybersecurity, the notion raises a preliminary caveat: unlike data protection, the emergence of a Brussels Effect in this domain cannot yet be empirically demonstrated. The regulation of the sector appears to be an ongoing process, and the EU Space Act has not entered into force. At this stage, the claim is therefore not that such an effect already exists, but rather that the Union appears to be assembling the legal and institutional conditions that could make it possible. As shown by Bradford, even though a country's market size is a good indicator of its ability to extend its regulatory reach, such influence does not stem from market size alone, but rather from its combination with said country's regulatory capacity and political will to do so.123 Therefore, it appears worthy to examine whether the normative and political conditions could be met for such an effect to emerge.

As for the material conditions, out of the five cumulative conditions necessary for the Brussels Effect to materialise, that is, a large and attractive

120 Ian Manners, 'Normative Power Europe: A Contradiction in Terms?' (2002) 40 Journal of Common Market Studies 235.

121 Marise Cremona and Joan Scott, *EU Law Beyond EU Borders: The Extraterritorial Reach of EU Law* (Oxford University Press 2019).

122 Christopher Kuner, 'The Internet and the Global Reach of EU Law' in Cremona and Scott (n 124).

123 Bradford (n 122).

internal market, sufficient regulatory capacity, political will to deploy stringent regulation, inelastic targets, and non-divisibility of production, the Union satisfies three without significant difficulty.[124] Its internal market represents a major destination for space services and products; its regulatory capacity in both cybersecurity and space governance is well established and expanding; and its political will to deploy stringent regulation in this domain is explicitly articulated in its own strategic documents (see Section II/ 2)). The two remaining conditions warrant more careful examination in the space context. Space operators are not wholly inelastic, unlike individual consumers, they can, in principle, configure separate systems for different regulatory environments. However, several features of the space sector limit this mobility such as the high costs of developing parallel system architectures, the technical interdependence of space and ground segments, and the economies of scale inherent in satellite manufacturing, making true divisibility economically untenable for most operators.[125] Moreover, access to EU-funded programmes, frequencies, and procurement markets constitutes an inelastic dependency that cannot easily be replicated elsewhere, as will be discussed later. Non-divisibility operates in that case through economic and institutional structure as operators integrated into European supply chains, programme ecosystems, and market access frameworks face compliance costs that make maintaining separate regulatory postures impractical.

First, several structural features of the space sector make it particularly receptive to this type of regulatory projection. Such an effect seems most

---

124 Bradford (n 122).

125 On the capital intensity and scale economies of satellite manufacturing, see Giacomo Gatto and others, 'Space industry trends: How Europe could outcompete' (*McKinsey*, 2025) <https://www.mckinsey.com/industries/aerospace-and-defense/our-insights/is-europe-still-on-the-launchpad-reshaping-its-space-ecosystem-to-lead#/> accessed 19 December 2025 ; on the technical interdependence of space and ground segments as a structural feature of space systems, see Bace and others (n 10) 3; on the economics of the New Space industrial model more broadly, see Punnala and others (n 10).

visible through three channels: market access, programme participation, and the industry-standardisation ecosystem for industry actors. Indeed, space infrastructure is deeply embedded in global supply chains; satellite systems rely on complex assemblages of components, software, launch services, and ground infrastructure sourced across multiple jurisdictions. At the same time, access to key markets, frequencies, or public procurement programmes is highly concentrated and politically structured. Thus, for the Union's markets, the relevance of these channels does not merely lie in the formal existence of EU requirements, but in the structural dependence of space industry actors on continued access to legally and institutionally central ecosystems. In a sector characterised by high entry costs, long development cycles, and strong reliance on public procurement and interoperability, exclusion from EU markets or programmes would entail more than the loss of isolated commercial opportunities; it would entail durable marginalisation from key segments of the global space economy.[126] It is this structural dependency, rather than empirical market dominance alone, that underpins the Union's capacity to induce regulatory alignment among foreign actors.

For instance, recent developments in the European launch sector further illustrate this dynamic of growing structural centrality. The successful commercial deployment, in February 2026, of 32 satellites from Amazon's LEO constellation by Ariane-6 under the largest launch contract in Arianespace's history reflects the Union's capacity to position its infrastructure as an indispensable node within global space value chains, including for non-European mega-constellations traditionally associated with US-based systems such as Starlink.[127] It appears that the Union is in the

[126] Sara Dalledonne, Justine Dousset and Marco Aliberti, 'New Challenges for European Space Industrial Policy' (2025) Global Policy.

[127] Pierre Manière, 'Ariane-6 réussit son premier lancement commercial pour la constellation de satellites d'Amazon' (*Le Monde*, 12 February 2026) <https://www.lemonde.fr/economie/article/2026/02/12/ariane-6-reussit-son-premier-lancement-commercial-pour-la-constellation-de-satellites-d-amazon_6666549_3234.html> accessed 12 February 2026.

process of re-establishing itself as a competitive provider of access to orbit through Arianespace and the Ariane-6 programme. The significance of this example extends beyond the commercial as the reestablished orbit access reinforced the credibility of regulatory conditionality attached to market access, programme participation and industrial cooperation. This growing infrastructural relevance for non-EU actors dependent on access to European launch services and programmes strengthens the Union's ability to induce compliance with its normative frameworks, including in the field of space cybersecurity.

In such a context, regulatory requirements imposed at critical nodes of the value chain, such as certification, licensing, market access, or public funding, can exert a powerful structuring effect well beyond the regulator's territory.[128] Thus, this domain offers fertile ground for observing the EU's attempt to diffuse its regulatory model beyond its borders. The EU's space cybersecurity initiatives, though territorially grounded, allow for a global reach; making space an instructive case study for examining the external dimension of EU norm-setting. Thus, this regulatory activity might reflect more than just a legal strategy; rather, it embodies the Union's broader identity as a normative power, as the EU uses internal regulatory sophistication to generate external influence.

It connects to a larger question about the EU's mode of international influence, and specifically about the relationship between its normative power identity and its market-based influence. Manners' theory of normative power describes the EU as an actor that shapes international behaviour through the diffusion of norms, values, and conceptions of appropriate conduct.[129] The Brussels Effect, as described above, differs from normative power which operates through persuasion and legitimacy, while

[128] On the value chain of the space economy see Danielle Binci, Andrea Appolloni and Wenjuan Cheng, 'Space Economy, Innovation and Sustainability' [2022] Symphonya. Emerging Issues in Management 110.

[129] Manners (n 123).

the Brussels Effect operates through structural dependency and compliance incentives. In the space cybersecurity context, both mechanisms appear to be at work, through different channels and with different implications. The Brussels Effect, through market access conditions, programme participation, and supply chain integration, as previously discussed, and the Union's normative power, emerging from its deliberate framing of its regulatory project as a contribution to international governance, as reflected in its institutional discourse and strategic documents.

The relationship between the two is thus complementary as the Brussels Effect provides the market mechanism through which EU norms spread, while normative power provides the political legitimacy and identity-driven framing that shapes how that diffusion is justified and received. Thus, this article treats them as analytically distinct but mutually reinforcing dimensions of the EU's external normative influence in the space cybersecurity domain. The diffusion of European standards beyond its borders, often through market access conditions or international alignment efforts, as discussed below, might lead to a form of Brussels Effect that would further the EU's capacity to set global norms unilaterally. In doing so, those normative strategies would reflect the EU's deeper identity as a normative power, shaping international influence by defining appropriate standards of behaviour and seeking to establish an ethical baseline for conduct in space activities.[130]

This external resonance of the EU's cybersecurity regulations seems particularly evident in the space sector, where both proposed and existing frameworks extend their influence beyond EU borders. The proposal for an EU Space Act illustrates this trend, as its resilience and cybersecurity requirements are intended to apply across entire supply chains, ensuring risk management throughout the full operational lifecycle of space systems, rather than being limited to specific phases or actors (see section I/2). In practice, European norms would function as de facto market access

[130] Manners (n 123).

conditions: manufacturers and operators outside the EU must comply with these uniform requirements if they wish to engage with the European market or participate in EU-funded programmes.

As noted in the literature, the ambition to extend the applicability of EU space regulation to non-EU operators was articulated well before the publication of any legislative proposal, through policy documents, stakeholder consultations, and institutional discourse that explicitly framed the EU Space Act as a tool for international norm-setting and market conditioning.[131] Indeed, as Commissioner Kubilius has noted, the Act lays 'the foundations for an EU space single market for space' and for 'trade with like-minded partners based on reciprocity', thereby linking regulatory alignment to market access and participation in the European space ecosystem.[132]

This regulatory export is not new, however. Even before the proposed Space Act, existing instruments such as the NIS2 Directive and the CRA had already begun to generate extraterritorial effects by imposing cybersecurity-by-design principles on digital products and critical infrastructure.133 Collectively, these frameworks reflect a broader EU strategy that mirrors its approach in the digital domain, where regulations, most notably the GDPR, have successfully projected European norms globally. In the emerging field of space cybersecurity, the Union appears to be replicating this approach, using its internal regulatory authority to shape practices well beyond its borders. Such a parallel with the GDPR is not merely a figure of speech.

---

131 Ingo Baumann and Katharina Prall, 'The EU Space Act' s Global Reach: Legal Considerations on the Envisaged Application to Non - EU Operators' (2025) Global Policy.

132 Commissioner Kubilius, 'Opening Address by Commissioner Kubilius at the 18th European Space Conference' (*European Commission*, 27 January 2026) <https://ec.europa.eu/commission/presscorner/detail/en/speech_26_228> accessed 28 January 2026.

133 Benjamin Farrand, Helena Carrapico and Aleksei Turobov, 'The New Geopolitics of EU Cybersecurity: Security, Economy and Sovereignty' (2024) 100 International Affairs 2379.

European institutional actors explicitly claim it themselves. In April 2024, the Director of EUSPA, Rodrigo da Costa, drew a direct analogy between the GDPR and the future Space Act, expressing the ambition that a similar global effect should occur in the fields of space infrastructure sustainability and cybersecurity.[134]

Industry actors have also voiced converging expectations. Notably, ASD Eurospace has called for the Space Act to apply to non-European operators to ensure that access to the EU market is conditioned on compliance with European standards, stating that 'EU Space Law shall apply to non-European stakeholders willing to address the EU market, in order not to create any competitiveness distortion with the European space industry'.[135] Taken together, these institutional and industry positions suggest that the extraterritorial reach of the Space Act is not an incidental feature but a deliberate regulatory strategy, in which the Union intends to use its internal market authority to shape practices and standards beyond its borders.

While a Brussels Effect does not necessarily result from extraterritorial regulations, nor does every instance of extraterritorial EU law generate a Brussels Effect, both dynamics remain closely intertwined as they contribute to the current global reach of the EU's space cybersecurity norms.136 As expressed in the proposed EU Space Act, the EU's extraterritorial reach is less about regulating foreign conduct per se than about preserving the integrity

---

134 Théophane Hartmann, 'Space Agency Director Thinks EU Space Law Would Have GDPR-Style Outsized Impact' (*Euractiv*, 25 April 2024) <https://www.euractiv.com/news/space-agency-director-thinks-eu-space-law-would-have-gdpr-style-outsized-impact/> accessed 16 August 2025.

135 ASD Eurospace, 'Position Paper on the Proposal for a Regulation on the Safety, Resilience and Sustainability of Space Activities in the Union (EU Space Act)' (Eurospace, November 2025) <https://eurospace.org/eurospace-position-paper-eu-space-act/> accessed 5 December 2025.

136 Austen Parrish, Cédric Ryngaert and Régis Bismuth, 'The European Union Experience of Extraterritoriality: When a (Willing) Victim Has Become a (Soft) Perpetrator', *Research Handbook on Extraterritoriality in International Law* (Edward Elgar Publishing 2023).

and competitiveness of its internal market.[137] By broadening the scope of application of its laws, under Article 114 TFEU, the EU aims to ensure a level playing field between EU and non-EU companies, thus protecting its internal economic balance. Therefore, the inclusion of entire supply chains within the scope of the proposed EU Space Act, for example, appears to be a defensive regulatory move aimed at preventing vulnerabilities and distortions of competition. Regarding this extraterritorial reach of EU regulation, the effects might in fact rarely be aimed at projecting influence per se, as they typically result from the internal objective of protecting EU interests through regulations that, incidentally or deliberately, extend their scope beyond borders.[138] Yet they appear to reinforce the EU's global reach. Both dynamics converge in their practical effect, that it, extending the reach of EU space cybersecurity norms beyond its territory.

That said, the EU seems to be conscientiously applying its internal norms to exert global influence, and there appears to be a growing tendency to justify extraterritorial application of those norms by framing the issue as falling within its territory.[139] This dynamic appears to go beyond a mere byproduct of internal objectives, as the use of certification schemes and standard-setting in space cybersecurity suggests a more deliberate deployment of EU normative influence. For example, the mechanism envisaged in the EU Space Act, under which non-EU operators may be presumed to meet EU cybersecurity requirements if the Commission issues an equivalence decision (Articles 16 and 105), mirrors the EU's established 'carrot and stick' approach in fields such as data protection and trade. As with adequacy decisions under

137 Parrish, Ryngaert and Bismuth (n 139).

138 Léna Hornkohl, 'The Extraterritorial Application of Statutes and Regulations in EU Law' (2022) SSRN Electronic Journal <https://www.ssrn.com/abstract=4036688> accessed 23 July 2025.

139 The EU's use of extraterritoriality is typically grounded in the protection of specific and subjective Union interests, such as safeguarding the internal market or the fundamental rights of its citizens. This is usually justified by a rational connection between the triggering factor (eg, effects doctrine or targeting criterion) and the EU's interest at stake, as seen in areas like competition or data protection law.

the GDPR, this method offers preferential access (the carrot) to third-country actors aligning with EU standards, while implicitly withholding market benefits from those that do not (the stick), thereby incentivising regulatory convergence.[140]

Those external effects created by binding regulatory requirements appear to be reinforced through non-binding mechanisms, contributing to what the article would describe as a 'compliance-as-norm projection' dynamic. In practice, access to the EU space market is becoming conditioned on adherence to European standards, encouraging non-EU firms to align with these norms to secure market entry. This fosters an additional form of soft extraterritorial reach, where regulatory influence extends beyond the Union's borders through voluntary alignment driven by commercial incentives rather than direct legal imposition. However, the scope of this soft normative projection remains constrained, as its effectiveness depends on private actors' willingness to comply, potentially leading to fragmentation if compliance varies across supply chains.

Taken together, these elements might be examined through the lens of the Brussels Effect, which holds that the European Union's internal regulatory choices exert influence beyond its borders. The development of the IRIS² constellation provides an illustration of how participation in EU programmes operates as a vector of normative projection in the space sector. Although data protection is not a space-specific constraint, the architecture of IRIS² has been significantly shaped by the need to ensure GDPR compliance by design, to the point of delaying deployment.[141] Such an example is instructive as the data protection obligations are an established case of the Brussels Effect.[142] Its weight is felt here within a space programme

140 Kuner (n 125).

141 'EU Accepts 'Best-and-Final Offer' Bid for Satellite Mega-Constellation Project' (*EuroNews*, 1 November 2024), <https://www.euronews.com/next/2024/11/01/eu-accepts-best-and-final-offer-bid-for-satellite-mega-constellation-project> accessed 16 January 2026.

142 Bygrave (n 70).

itself, demonstrating that access to EU flagship programmes is effectively conditioned on the internalisation of EU legal and value-based requirements into the technical design of space systems.[143] EU requirements, therefore, function as de facto filters regarding the access to the European Space ecosystem, a logic that the future Space Act could extend and systematise in the field of space cybersecurity and resilience. By analogy, by embedding cybersecurity requirements into supply chains, certification processes, and market access conditions, the EU could be constructing the conditions for such an export of its space cybersecurity norms to non-EU actors.

Although the credibility of the Union's capacity to project its regulatory and cybersecurity norms beyond its borders depends on the structural position it occupies within the space sector, it appears that its position as one of the main space powers could assure its space industry and market enough relevance to see some global diffusion of the norms that condition its access.[144] Indeed, the EU is not a marginal actor, as '[d]espite its modest workforce and output, the European satellite industry exerts a strong leverage effect on EU trade'.[145] In Europe, public spending remains the primary market driver.[146] At the same time, private investment remains comparatively limited and concentrated in early-stage projects, reinforcing the centrality of institutional programmes for industrial survival and scaling.[147] Thus, through flagship programmes such as Galileo, Copernicus and IRIS², the Union has become one of the world's major institutional

---

143 Marco Franzoso, 'The 'Third Way' to Space Power: Europe's Digital Sovereignty Advantage' (*Stanford Law School*, 2026) <https://law.stanford.edu/2026/01/23/the-third-way-to-space-power-europes-digital-sovereignty-advantage/> accessed 18 February 2026.

144 Isabelle Sourbes Verger, 'La place de l'Europe au sein des puissances spatiales', in *L'espace extra-atmosphérique et le droit international* (2021) <https://hal.science/hal-03513360> accessed 16 May 2024.

145 Dalledonne, Dousset and Aliberti (n 129).

146 ESA, 'Report on the Space economy 2024' (*ESA*, December 2024) <https://space-economy.esa.int/documents-area > accessed 5 January 2026.

147 McKinsey (n 128).

customers, infrastructure operators and organisers of demand for space services. In space economy, a sector that remains overwhelmingly driven by public procurement and long-term programmes rather than by consumer markets, such a centrality gives the Union significant leverage.[148]

In such a technical, capital-intensive and highly integrated sector, exclusion from EU frameworks does not merely imply the loss of a commercial opportunity. Still, it risks durable marginalisation from core industrial ecosystems, supply chains and space applications.[149] Access to European programmes can thus be seen as a gatekeeping mechanism, that is, compliance with EU technical and security requirements becomes a precondition for participation in the space ecosystem. As a result, even non-EU manufacturers, operators and service providers face strong structural incentives to align with EU technical, security and compliance requirements to remain eligible for participation in these programmes and the industrial ecosystems they structure.[150]

This structural dynamic extends to the regulatory plane. Satellite operators and manufacturers based outside the Union may be compelled to comply with EU cybersecurity norms, whether through adoption of technical standards, conformity assessments, or product certifications, to access the internal market of the Union or participate in European space programmes. In this sense, the Space Act's extraterritorial ambition would rest on the expression of a structural reality, as the EU's position as a major organiser of demand in the space economy gives its regulatory requirements a pull that extends beyond its formal jurisdictional boundaries.

Overall, the external reach of EU space cybersecurity regulations reflects its broader capacity to convert market size into legal influence. This regulatory

148 ASD Eurospace, 'The European space industry in 2021' (*ASD Eurospace*, 2022) <https://www.asd-europe.org/news-media/publications/asd-reports-publications/facts-figures-2021/> accessed 17 June 2024.

149 McKinsey (n 128).

150 Baumann and Prall (n 134).

power depends not only on the Union's market attractiveness but also on its regulatory capacity, specifically its ability to identify and formulate relevant rules, provide clear and coherent normative frameworks, and ensure effective enforcement.[151] Much as with the GDPR, the EU positions itself as a regulatory model for cybersecurity of space infrastructure, using its internal norms to shape external market expectations, some already in place such as with NIS 2 Directive or the CRA and some under negotiation, notably the proposed Space Act.[152] In this context, compliance with EU requirements functions as a form of normative soft power. While a de facto Brussels Effect cannot yet be empirically demonstrated in this domain; given that the central instrument, the Space Act, has not entered into force; identifying the legal and structural conditions under which such an effect could emerge is itself analytically valuable, particularly in a field where regulatory choices made now will shape the governance architecture for decades.[153] Indeed, rather than imposing direct regulatory dominance, as seen with instruments such as the US ITAR, the EU promotes the diffusion of its norms through supply chains and market dependencies. For many non-EU actors, especially emerging space operators, aligning with established EU norms might then often appear more practical than developing their own, enabling the Union's rules to spread beyond its immediate jurisdiction through a logic of regulatory adoption.[154]

---

[151] EU's ability to transform its market size into legal influence depends on its regulatory capacity, which in turn relies on its expertise (identification and formulation of rules), its coherence (clear and predictable normative frameworks), and its enforcement power (ability to ensure compliance with the standards). See David Bach and Abraham Newman, 'The European Regulatory State and Global Public Policy: Micro-Institutions, Macro-Influence' (2007) 14 Journal of European Public Policy 827.

[152] Abraham Newman , *Protectors of Privacy* (Cornell University Press 2008); Bradford (n 122).

[153] Manners (n 123) ; The *de facto* Brussels Effect refers to non-EU companies aligning with EU standards to access the EU market and subsequently applying them globally, while the *de jure* version involves third countries voluntarily incorporating EU rules into their domestic law. Both describe regulatory influence beyond borders., Hornkohl (n 141).

[154] Kuner (n 125).

In sum, the progressive construction of its internal framework contributes to shaping the EU's emerging role as a normative power in the domain of space cybersecurity. Instruments such as the NIS2 Directive, the CRA, and the upcoming EU Space Act not only regulate the European market but also reflect a broader strategic ambition to influence global practices. By exporting its regulatory logic, the Union is positioning itself as a standard-setter in an increasingly contested technological domain, in the absence of binding international law. This strategy seeks to position the Union not only as a standard-setter, but as one of the architects of international regulation through normative projection. While its influence may depend on voluntary alignment, the interplay of binding and non-binding tools appears to reinforce its capacity to structure behaviours beyond its borders, as analysed in Section I/ 2) and above.

That said, important limits must be acknowledged. The effectiveness of this strategy will depend on several contingent factors: the actual stringency and coherence of the final Space Act, the credibility of its enforcement, the relative attractiveness of the EU market and programmes such as IRIS², and the reactions of competing regulatory powers, notably the United States.[155] Unlike data protection, where the EU regulates a near-unavoidable global market, space activities remain more geopolitically fragmented and strategically contested.[156] Whether this strategy will ultimately lead to a genuine Brussels Effect remains to be seen. Still, the legal and institutional conditions for such an outcome appear to be in place.

---

[155] For instance, the U.S. Government voiced its concerns over the intended scope of the EU Space Act, estimating that it would impose 'unacceptable regulatory burdens on U.S. providers of space services to European customers' and therefore requesting the text to be amended so as to the ensure the EU 'proceed cautiously' under threat of ceasing the cooperation between the two space powers; Office of Space Commerce, 'U.S. Government Provides Feedback on Draft E.U. Space Act' (*Office of Space Commerce*, 20 November 2025) <https://space.commerce.gov/u-s-government-provides-feedback-on-draft-e-u-space-act/> accessed 4 December 2025.

[156] Florence Gaillard-Sborowsky, *À la recherche d'une sécurité spatiale.* (Le Cavalier Bleu, 2023).

2. From strategic regulation to regional norm leadership: the consolidation of a European governance layer

Rather than simply producing new rules for space cybersecurity, it can be argued that the EU is actively constructing a new governance layer at the regional level, one that fills the international legal vacuum described earlier. This regional framework does not purport to bind non-EU states directly, but rather to establish a credible normative reference point that may, over time, exert influence beyond EU borders through the dynamics analysed in the preceding section. In doing do, it positions the EU as a regional normative leader in a blocked multilateral context, framing the cybersecurity of space as both a normative and strategic priority. Even more so as it could be seen as 'strategic legislation', that is, law-making that is explicitly aligned with broader political, economic and technological objectives.[157] Such a consolidated governance layer would then help make the Union a 'strategic actor in its own right' in space.[158]

Applied to space cybersecurity, this approach would go beyond the adoption of isolated security requirements, with its emerging layered and increasingly coherent governance architecture, combining horizontal instruments such as NIS2 and CER Directive, sector-specific regulation, notably the proposed EU Space Act, standardisation strategies, certification schemes, supervisory structures and programme-based conditionalities. Taken together, these elements hint at a new regulatory layer that does not merely regulate behaviour *ex post*, but organises the space sector around risk management, compliance, auditability and resilience-by-design. This development seems to mark a transformation in the way space activities are framed, as the emerging European approach 'cyberises' space governance. That is, it

[157] 'In the context of the European Union, strategic legislation can be defined as "A set of laws and regulations aligned with the EU's overall strategy and aimed at achieving planned goals. Strategic legislation benefits stakeholders and promotes cooperation within the internal market', Cappelletti and Papakonstantinou (n 26) 5.

[158] Philip De Man and Jan Wouters, 'EU Space Governance at the Threshold of A New Era' (2025) Global Policy.

replaces *ex post* responsibility with *ex ante* risk management, legal attribution with continuous compliance, and abstract obligations with operational requirements such as incident reporting, penetration testing, supply chain security, and governance by audit. This development marks a transformation in the framing of cybersecurity obligation in the space sector at the European level – as a preventive layer is being constructed.

While the lack of binding international legal instruments in space cybersecurity has been acknowledged earlier, what sets the EU's approach apart is the way it is constructing a progressively layered legal framework to fill the perceived gap.[159] Identifying a regulatory gap, the EU is responding not by waiting for multilateral consensus but by developing its own internal framework, with layered, hybrid, and increasingly sector-specific initiatives.[160] The proposed EU Space Act illustrates this shift by building on broader instruments such as the NIS2 Directive and the CER Directive, yet is explicitly framed as *lex specialis*, marking a key moment in the Union's move from general cybersecurity regulation to tailored legal frameworks for space systems. This nested regulatory architecture, where sectoral laws are embedded within horizontal ones, allows the EU to progressively specialise and raise baseline standards for resilience across all segments of space infrastructure, including national and private operators. In the absence of

[159] Indeed, EU's own communication point out an issue as existing international instruments, particularly the Outer Space Treaty, remain fundamentally state-centric and technologically outdated, leaving gaps on some issues. As exposed by the proposal for an EU Space Act, those fundamental texts do not provide detailed responses to today's critical challenges, including cyberattacks on space infrastructure, the environmental impact of space activities, or the congestion of orbital zones and multilateral instruments, such as the UN's Long-Term Sustainability Guidelines, offer only non-binding recommendations, leaving a growing set of risks legally unaddressed. See EU Space Act (n 32) 15.

[160] Indeed, in the explanatory memorandum for the proposed EU Space Act, the "lack of specific binding technical rules to implement the general OST obligations" is proposed as a reason for a 'fragmented regulatory landscape' that must be remedied by the adoption of said Act.

international law, the EU seems to be constructing its own norms by adding a layer of specialised sectoral regulation.

For instance, the soft law approach to space cybersecurity appears not only strategic but also intentionally pragmatic, functioning as a complement to the existing international legal framework - that is, the general instruments of space law that do not contain preventive and specific cybersecurity obligations (see Section I/ 1). Centred on the formulation of technical standards, particularly aimed at enhancing the security of LEO and GEO constellations, this soft-law-based approach positions the EU as a normative frontrunner in protecting and enhancing the resilience of space infrastructure.[161] Framed as a pragmatic response to multilateral inertia, it relies on the diffusion of guidelines and best practices rather than binding commitments. Presented as a provisional mechanism 'in anticipation' of future hard law, the use of non-binding standards reflects a broader strategy of normative soft power.[162] However, such instruments face limitations in achieving international alignment, particularly amid divergent regulatory models and geopolitical competition among major space powers.[163] As such,

---

[161] See, for example, the progress envisioned in the 'European Connectivity Programme', Clément Evroux , 'EU Secure Connectivity Programme 2023-2027' (European Parliamentary Research Service, February 2023) <https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2022)729442> accessed 20 January 2025; see also, the exposition of the Union's classification of space systems as critical infrastructure being still under discussion by other major space powers, Casaril and Tricco (n 72) .

[162] Manners (n 123).

[163] Katrina Moon, Mark Glissman and Allison Dempsey, 'Space Traffic Management' (2025) 12 Journal of Space Safety Engineering 327 ; see also Michael Gleason, 'Establishing Space Traffic Management Standards, Guidelines and Best Practices' (2020) 7 Journal of Space Safety Engineering 426. <https://www.sciencedirect.com/science/article/pii/S2468896725000242> accessed 5 May 2025; and Kuan Yang, 'Chinese Perspective on an International Regime of Space Traffic Management' (2019) 6 Journal of Space Safety Engineering 156 <https://www.sciencedirect.com/science/article/pii/S2468896719300631> accessed 5 May 2025; Jie Long and Tao Zhang, 'Pillars of Space Traffic Management in the Era of

the EU's normative ambitions, while often cited as a reference point, remain inherently fragile and open to global contestation.

Despite this fragility, the EU strategy appears to be showing some results, largely due to stakeholder involvement and the structured diffusion of standards. The example of ECSS's central role in harmonising technical practices across the European space sector illustrates such effectiveness. In practice, ECSS standards increasingly serve as technical reference points within research.[164]

Additionally, while ECSS standards are not imposed at the global level, they increasingly participate in international standardisation processes, notably through formalised cooperation and harmonisation efforts with global bodies such as ISO.[165] By striving for technical compatibility and complementarity, particularly within the ISO Technical Committee for Space Systems, ECSS contributes to the convergence of European and international space standards. The Commission's guidance also explicitly couples European space standards with international norms; for instance, while component qualification is framed by ECSS standards, core evaluation metrics, such as TRLs, are defined by direct reference to ISO Standard 16290.[166] Finally, ECSS standards are frequently mandated by the ESA, which is basing its list of Approved Standards (LEAS) applicable to ESA space

LEO Mega-Constellations: A Global Perspective' (2024) 74 Advances in Space Research 800 <https://www.sciencedirect.com/science/article/pii/S0273117724003430> accessed 27 March 2025.

164 See for example Ricardo Eito-Brun, Juan Miguel Gómez-Berbís and Antonio de Amescua Seco, 'Knowledge Tools to Organise Software Engineering Data: Development and Validation of an Ontology Based on ECSS Standard' (2022) 70 Advances in Space Research 485.

165 Point of contact is for instance maintained with ISO in its Technical Committee (TC) 20, Subcommittee (SC) 14 Space Systems, for the purpose of harmonisation and complementary actions; for further details on the cooperation, see 'ISO – ECSS – European Cooperation for Space Standardization'.

166 European Commission (n 98) 6.

projects on published ECSS standards.[167] Such coupling illustrates the idea that the EU is, if not yet globally diffusing, at least clearly diffusing its standards in the regional layer.[168] Moreover, this mechanism would echo the dynamics observed in digital regulation, where the Brussels Effect enables European standards to spread beyond the EU's borders, as companies comply with them to access the European market, as both the structural parallels drew earlier and the Union's stated ambition converge.[169]

Indeed, while this regulatory strategy is not novel for the EU, it is being applied in the space sector in a targeted and increasingly structured manner.[170] Confronted with the growing influence of China within international standard-setting bodies, where technical standards are directly linked to geopolitical and economic interests, the EU has made it a priority to assert its presence in pre-normalisation and standardisation processes.[171]

---

167 The reference to a objectives of implementing ECSS standards can be found on ESA's page, ESA , 'Requirements and Standards' <https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Requirements_and_standards> accessed 14 January 2026; see also Werner Kriedte and Youssef El Gammal, 'A New Approach to European Space Standards' [1995] ESA Bulletin.

168 The regional layer here is understood as the fact that the EU is at least achieving consistent uptake across the European space sector, encompassing EU Member States, ESA member states, and the European industrial ecosystem more broadly. This regional consolidation establishes a critical mass of adoption that would be the precondition for any kind of outward diffusion, as standards that are uniformly applied within a large and technically sophisticated regional market tend to become reference points for actors seeking access to it.

169 Newman (n 133); and Bradford (n 122).

170 This regulatory strategy of using standardization as a tool of normative projection is fitting within previous broader and well-established EU approach, as evidenced by previous initiatives such as the GDPR. See Kuner (n 125).

171 See for example China's "standards 2035" in the "National Standardization Development Outline", '中共中央 国务院印发《国家标准化发展纲要_新华网' ['The Central Committee of the Communist Party of China and the State Council Issue the National Standardization Development Outline'] (新华网, 10 October 2021) <http://www.news.cn/politics/zywj/2021-10/10/c_1127943309.htm> accessed 7 November 2025.

This general strategic posture of competing for normative influence in standard-setting as a matter of geopolitical priority provides the context within which the EU's more specific ambitions in space cybersecurity standardisation should be understood. The space sector is one instance of a wider competition, and the EU's early positioning in space cybersecurity standard-setting reflects the same logic applied to a domain where the race is still open. This ambition is clearly articulated in the renewed European Standardisation Strategy, which identifies space as a critical sector and aims to develop technical specifications specific to the European space economy, first to recognise them as European standards, then as international standards.[172] By coordinating Member States, national standardisation bodies, and specialised agencies, the EU is embedding space standardisation within a coherent strategic framework that explicitly treats technical standards as instruments of influence. This approach not only enhances the Union's capacity to shape global norms in line with its interests and values but also strengthens the international competitiveness of its space industry.[173] A position that the proposed EU Space Act aims to consolidate by establishing standardisation as both a tool of governance and an instrument of strategic autonomy, one that space matters increasingly seems to enable.[174]

---

[172] European Commission, 'Communication for the Commission to the European Parliament, the Council, the European economic and social committee and the committee of the regions: An EU Strategy on Standardisation Setting global standards in support of a resilient, green and digital EU single market' (2022) COM (2022) 31 final.

[173] European Commission, 'Communication for the Commission to the European Parliament, the Council, the European economic and social committee and the committee of the regions: A vision for the European Space Economy', (2025) COM (2025) 336 final.

[174] The concept of 'strategic autonomy' appears to be increasingly exploited both by commentators and by the Union's own discourses, and is mostly used as both a descriptor of a state of non-reliance on outside provider of strategic means (those ranging from materials, such as microchips, to intangible alliances, exemplified for example by the idea of a 'Defense Europe') and as an objective to achieve to further the EU's position as a global actor. See on this concept: Niklas Helwig and Ville Sinkkonen, 'Strategic Autonomy and the EU as a Global Actor: The Evolution, Debate and Theory of a Contested Term' (2022)

Moreover, amid persistent deadlocks in multilateral negotiations, the European Union appears to embrace a pragmatic regionalist strategy for space governance. Rather than waiting for a global consensus to emerge, the EU seeks first to consolidate credible and effective standards at the European level, then to promote their international uptake.[175] This two-step logic is not new. It underpinned the success of the GDPR in digital governance and is now resurfacing in the space domain, particularly in areas such as STM, where EU standards already exert extraterritorial effects.[176] This approach relies on a spillover effect as once a set of standards is broadly applied within the internal market, it often becomes a *de facto* global benchmark, particularly for international operators seeking access to the European market. By creating legal obligations that extend across supply chains and market access requirements, the Union effectively projects its standards outward. This governance model aims not only to fill a void but to shape what comes next, filling the legal vacuum with regional rules that may later act as global precedents.

While vulnerable to fragmentation and competing visions, it creates a concrete, strategic response to legal inertia at the multilateral level, and it signals the EU's intention to lead, at least regionally, on questions of security, resilience, and sustainability in space. By presenting a regional model of

27 European Foreign Affairs Review 1; Chiara Cellerino, 'EU Space Policy and Strategic Autonomy: Tackling Legal Complexities in the Enhancement of the 'Security and Defence Dimension of the Union in Space' 2023 8 European Papers - A Journal on Law and Integration 487 <https://www.europeanpapers.eu/en/europeanforum/eu-space-policy-and-strategic-autonomy> accessed 29 November 2024; see on the subject of space as an enabler of the Union's strategic autonomy: Letizia Caito, 'European Technological Non-Dependence in Space' (2015) 51 ESPI Report; Daniel Fiott, 'The European Space Sector as an Enabler of EU Strategic Autonomy' (Policy Department for External Relations - Directorate General for External Policies of the Union, 2020).

175 Mario Barbano, 'Space Traffic Management and Space Situational Awareness: The EU Perspective' (2022) 47 Air and Space Law 451.

176 European Commission, 'Joint communication to the European Parliament and the Council: An EU Approach for Space Traffic Management An EU Contribution Addressing a Global Challenge' (2022) JOIN (2022) 4 final, 13.

normative leadership in the absence of global consensus, the EU appears to follow a strategy of proposing norms at its level, in the hope that they spread. Regionalism thus seems to be a reasonable response to the blockage of multilateralism, in which the EU's governance model could encourage upward convergence towards the multilateral adoption of international norms.

This approach reveals a pragmatic dual objective: regional consolidation is the immediate goal, while international norm projection is a secondary, implicit ambition. A similar dynamic was observed in the 2010s, when the launch of the Union's Space Situational Awareness programme coincided with its efforts to advance a Code of Conduct for Outer Space Activities, reflecting its first attempts to secure critical space infrastructure.[177] Faced with the absence of binding multilateral instruments and the increasing fragmentation of space governance, the EU acts first out of necessity to protect its infrastructure, secure its internal market, and address legal uncertainties left by outdated treaties. This explains the progressive layering of sector-specific instruments, notably the proposed EU Space Act, over broader cybersecurity frameworks, for example, the NIS2 Directive. In space cybersecurity, the EU may not formally articulate norm export as a primary goal, but it nonetheless seems to recognise the geopolitical potential of regulatory leadership.[178] Through its control of standards, certification procedures, and compliance mechanisms, often diffused through market access and partnerships, the Union shapes global practices. This dual movement reflects not only the dynamics of the Brussels Effect but also a

[177] Phillip Slann, 'Anticipating Uncertainty: The Security of European Critical Outer Space Infrastructures' (2016) 35 Space Policy 6.

[178] For instance, in its Space Strategy for Security and Defence, the Union frames space governance as a strategic normative project, stating that 'The main challenge for the EU […] is to convince the vast majority of UN member countries of the relevance of a normative approach', and that it will 'work towards broadening international support for their position on outer space'. European Commission, 'Joint Communication to the European Parliament and the Council European Union Space Strategy for Security and Defence' (2023) JOIN (2023) 9 final.

secondary strategy to assert normative power in a domain where global consensus remains elusive.[179]

Despite this ambition to take a leading role in regulating space cybersecurity, several challenges raise questions about the strategy's actual reach and effectiveness. The success of the governance layer that the EU is attempting to construct ultimately hinges on industry actors' uptake, particularly in the case of its non-binding initiatives.[180] Yet, industry adoption cannot be taken for granted. Private operators may resist regulatory frameworks perceived as overly complex or burdensome, especially if compliance entails high operational or financial costs.[181] As several commentators, including those from within the space industry and legal circles, have noted, regulation alone will not suffice: without adequate incentives, funding mechanisms, or support structures, the EU's leadership may remain more declaratory than transformative.[182] Critiques have also emerged around the perceived proactivity of the EU in areas where it is not formally a party to international treaties, raising questions about potential interference with Member states' work in international fora, especially the UNCOPUOS.[183] These tensions reflect deeper structural limits to the EU's governance model: while it can project norms through internal regulation, its success depends on whether those norms are accepted, implemented, and supported by European industry and international partners. This dependency on private actors is particularly significant at the international level as the governance of space cybersecurity notably relies on technical standards developed by private

[179] Deliberate identification with a normative power as concept can for example be found when the High Representative of the Union for Foreign Affairs and Security Policy described the Union as a 'key normative power in setting world-wide rules and standards'. Josep Borrell, 'A Stronger Europe in the World' (Hearing of Josep Borell High Representative of the Union for Foreign Affairs and Security Policy), Committee on Foreign Affairs of the European Parliament, Brussels, (2019) 6.

[180] Frison-Roche (n 117).

[181] ASD Eurospace (n 138).

[182] Pavur (n 5); Livingstone (n 13).

[183] Baumann and Prall (n 134).

standardisation bodies and implemented by private operators; meaning that industry uptake is a condition for the international credibility and diffusion of the EU's regulatory model.[184] Finally, even if said norms are far-reaching and adopted, their adoption by third countries may appear voluntary. Yet, the line between emulation and subtle coercion is often hard to draw. In some instances, alignment with EU norms may be driven less by independent policy choices than by implicit political pressure.[185]

Moreover, in pursuit of regional-level normative leadership, the EU remains constrained by its nature as an international organisation, bound by the letter of its foundational treaties and by the extent to which its Member States agree to delegate power to its institutions. Those deeper structural constraints, some inherent to the EU's nature as an international organisation and some reflecting genuine political tensions with Member States over the scope of delegated competence, are evident as, while the EU Space Act proposal is presented as a harmonising instrument, the very possibility of harmonisation in space matters is formally excluded by Article 189(2) TFEU. This provision clearly states that Union measures in the field of space policy must not lead to the harmonisation of Member States' laws and regulations. Although the Commission has grounded the proposed Space Act in Article 114 TFEU, which relates to the internal market, to sidestep this constraint, such a move is not immune to criticism.[186] Legal scholars and Member States

184 Ikitemur, Karabacak and Igonor (n 120).

185 Kuner (n 125).

186 Malin Wållgren, 'Exploring the Outer Limits of Article 114 TFEU — Towards a General Power? An Analysis of Non-Market Objectives and "Measures Having as Their Object the Establishment and Functioning of the Internal Market"' (Uppsala University, 2016); Bas Jacobs, 'Understanding the EU's Competence to Harmonise Space Law Amid Publication Delays' (2024) 49 Air and Space Law <https://kluwerlawonline.com/api/Product/CitationPDFURL?file=Journals\AILA\AILA2024030.pdf\> accessed 17 January 2026; Dimitra Stefoudi, 'EU Space Law — Three Reasons Against, Three Reasons in Favour' (EJIL: Talk!, 29 April 2024) <https://www.ejiltalk.org/eu-space-law-three-reasons-against-three-reasons-in-favour/\> accessed 14 November 2024.

alike might voice concerns that this legal basis would exceed the EU's competence, especially when the regulation touches on matters closely tied to national sovereignty and security, such as the issuance of launch authorisations or the operation of sensitive infrastructure. Therefore, the stated objective of 'harmonisation' embedded in the proposed EU Space Act is not without controversy. Some fear that it may override or marginalise national positions or create legal frictions within multilateral settings, particularly in areas such as space resource exploitation, where legal authority remains contested. The resulting ambiguity raises the question of whether the EU is overreaching or attempting a strategic reinterpretation of its role by innovating norms in an area traditionally governed by intergovernmental logic. In this sense, the Union's push for 'harmonisation' may paradoxically weaken its normative project, if it alienates Member States or creates legal uncertainty around the enforceability of its regulatory instruments.

Finally, while the Union often frames its regulatory approach as an expression of normative power, grounded in values of rule-based governance, market openness, and multilateral cooperation, its increasing reliance on regional regulation in the space sector raises some interesting questions.[187] In the absence of binding international legal frameworks adapted to emerging challenges, for instance, cybersecurity, the EU's use of internal legislation to fill the normative vacuum may also signal a shift away from traditional multilateralism. Rather than strengthening global legal consensus, it would thus risk fostering legal fragmentation, particularly if other space powers continue to adopt similar unilateral or regional approaches.[188] The Union's initiative, while covering a genuine gap, therefore risks contributing to the fragmentation it nominally seeks to address.

---

187 Manners (n 123).

188 On the increasing tendency to supplant, or at least supplement, international multilateral space law through national or regional normative structures, See Achilleas and Hobe (n 18) 21.

Moreover, the EU itself justifies its regulatory initiatives in explicitly strategic terms, emphasising the need to protect its autonomy, assert its place on the international scene, and ensure the competitiveness of its space sector.[189] This narrative, while understandable, paradoxically echoes the very logic of power competition that space law historically sought to temper. The construction of regional governance frameworks as substitutes for, or at least complements of, multilateral instruments reflects a broader transformation in the architecture of space law today. This reveals a growing reliance on inventive, sometimes asymmetric legal tools to regulate newer issues in domains such as cybersecurity or commercial exploitation, where international law remains vague or silent. While these innovations respond to urgent regulatory gaps, they also underscore the limits of consensus-based norm-making. They may, in the long term, reinforce a fragmented legal order shaped by the interests and capacities of dominant space actors.

In sum, the Union's normative initiatives in space cybersecurity aim to position it as a regulatory leader, offering a governance model in the absence of effective multilateral consensus. By developing layered, sector-specific regulations and projecting them beyond its borders, the EU appears to strive for both strategic autonomy and normative influence. In this approach, norm projection serves both as a consequence of market dynamics and as a strategic tool. However, this trajectory is not without friction as regional normative leadership ambitions may provoke resistance from other space powers, especially where interests diverge, and relies heavily on the engagement of private actors to ensure operational uptake. While the EU model offers structure in an otherwise fragmented landscape and may serve

[189] See notably Council of the European Union, 'Council conclusions on 'Strengthening Europe's competitiveness through space" (23 May 2024) ST 10142/24; General Secretariat, 'Strategic Autonomy, Strategic Choices' (Council of the European Union Issues paper, 5 February 2021). See also Fiott (n 177).

as a stepping stone toward broader international convergence, whether it can scale globally or truly catalyse multilateral revival remains to be seen.

Conclusion

The increasing complexity and vulnerability of space systems in the digital age, especially in LEO, have exposed major shortcomings in the current international legal framework. Against this backdrop, this paper has examined whether and how the European Union might be developing a distinct normative response to the cybersecurity of space infrastructure, and whether this response can be understood as more than a purely internal regulatory exercise. As this paper observed, the EU responded by positioning itself both as a regulator addressing internal-market and security concerns and as a strategic and normative actor, developing a creative governance approach to space cybersecurity. This approach, progressively layering soft and hard law with standardisation and compliance mechanisms, seems to position the EU as not only responding to internal security needs but as an actor capable of influencing the global direction of space governance, simultaneously serving its strategic interests and reflecting its broader identity as a normative power.

With the proposed EU Space Act, the shift from the ad hoc inclusion of space infrastructure in general cybersecurity frameworks to the development of a dedicated, sector-specific legal framework seems confirmed. This progression aligns with the Union's long-term commitment to consolidating the security of its space infrastructure while projecting its regulatory standards outward. Far from a purely technical exercise, the EU's normative strategy relies on a combination of hard law, soft law instruments, compliance mechanisms, and standard-setting, all of which create the conditions to extend the EU's influence beyond its borders, particularly by conditioning market access on compliance with EU-defined norms. This approach might reflect a pragmatic dual strategy. On the one hand, regional consolidation, motivated by the absence of binding international law,

remains the primary objective – and most effective level of action, where the Union seeks to coordinate internal practices, secure space assets, and close legal gaps. On the other hand, norm projection emerges as a secondary but strategic ambition, in which the EU leverages internal norms to shape global practices. Taken together, these elements suggest that a degree of international diffusion of EU norms may occur, supporting the hypothesis of a potential Brussels Effect in the field of space cybersecurity.

However, the effectiveness and sustainability of this strategy remain uncertain. The absence of international consensus leaves room for competing frameworks led by other global actors, who may seek to promote their own distinct models of governance and regulation. In such a competitive environment, the Union's ambition as a normative leader may ultimately be perceived as merely one regional approach among others, increasing the risk of regulatory fragmentation and norm rivalry across regulatory spaces. The broader implications of this analysis extend beyond the case of space cybersecurity. The Union's approach illustrates a shift in how international regulation is evolving in technically complex and politically charged domains where multilateral consensus remains elusive. The rise of regional regulatory areas might complement existing international law, but it might as well produce a fragmented legal order in which the applicable norms depend less on international consensus than on which regulatory power an operator is economically dependent on. Whether this constitutes a productive form of regulatory experimentation or a structural threat to the original ambitions of space law is a question that goes beyond this article's scope, but one that the EU's normative activity in space cybersecurity makes increasingly necessary to address.

Ultimately, the EU's normative response to space cybersecurity governance can be seen as part of a broader transformation of international law in the space domain, where regulatory inventiveness increasingly compensates for multilateral paralysis. Its ability to sustain this ambition of leadership and to transform a regional solution into a global reference framework will depend

not only on political cohesion and industrial alignment, but also on the international community's receptiveness. In this sense, the EU's normative approach to space cybersecurity should be understood less as an established model of global governance than as a laboratory of regulatory experimentation. Whether this regional model might globalise and lead to a fully effective Brussels Effect thus remains to be seen, but the more consequential question may be what it would mean for the architecture of international space law if it does.